

Portfolio Backtest Evidence: 1997-2026

Net USD evidence for every current portfolio output, regional index benchmarks, and an independent point-in-time index challenger.

The long-history portfolio results are retrospective holdings replays, not point-in-time stock-selection backtests. Use them for exposure and path diagnostics, not as evidence that the current model could have selected these names in 1997.

13 portfolio outputs tested	29.6 requested historical years	\$186,060,522 current-derived starting NAV	2013-09-30 common 80% investable start	25 separate reconstructed PIT months
19 standalone benchmark paths	13 source-backed macro-event windows			

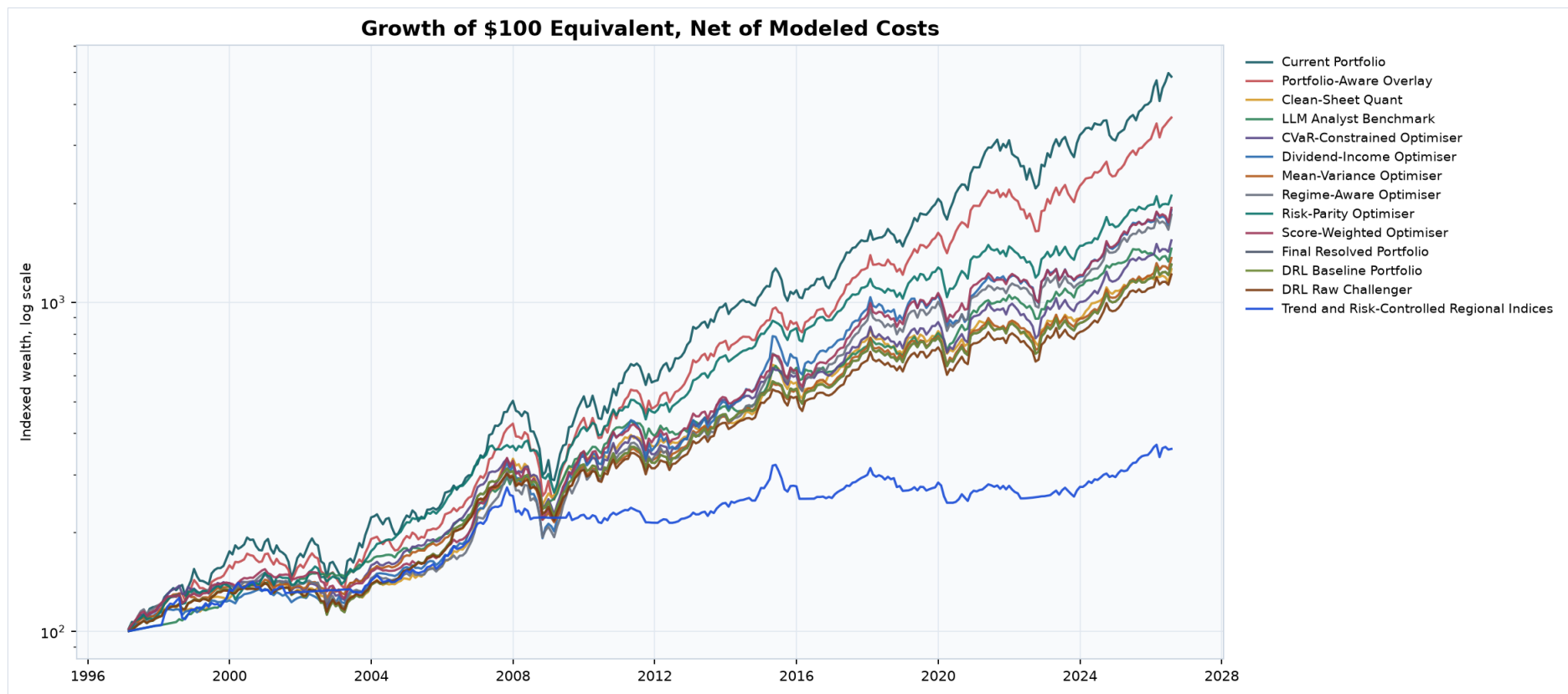
The test covers 29.6 years and 13 portfolio outputs. The current portfolio starts at \$186,060,522 and finishes at \$9,013,105,522 after modeled costs, a net PnL of \$8,827,045,000. On the common investable window, Portfolio-Aware Overlay has the highest Sharpe ratio at 0.88. The long results replay securities selected with today's information, so they describe exposure and path behavior rather than proving historical selection skill.

Executive Comparison

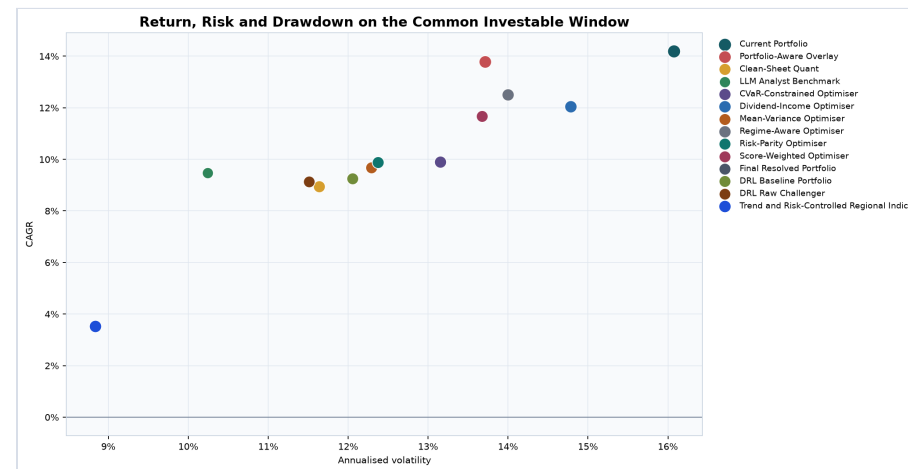
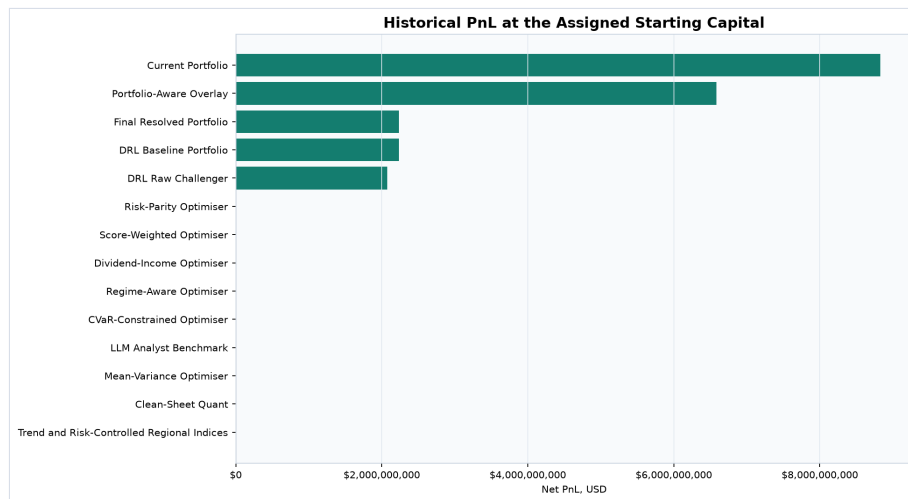
Portfolio	Start	cagr	Volatility	sharpe	Max drawdown	Ending value	PnL
LLM Analyst Benchmark	\$100,000	9.50%	9.07%	0.81	-20.91%	\$1,454,455	\$1,354,455
Portfolio-Aware Overlay	\$186,060,522	12.96%	14.43%	0.77	-41.21%	\$6,770,759,519	\$6,584,698,997

Portfolio	Start	cagr	Volatility	sharpe	Max drawdown	Ending value	PnL
Risk-Parity Optimiser	\$100,000	10.89%	11.54%	0.77	-30.56%	\$2,108,263	\$2,008,263
Current Portfolio	\$186,060,522	14.06%	16.66%	0.75	-42.57%	\$9,013,105,522	\$8,827,045,000
Score-Weighted Optimiser	\$100,000	10.57%	12.70%	0.69	-35.52%	\$1,935,381	\$1,835,381
Mean-Variance Optimiser	\$100,000	9.26%	11.01%	0.66	-29.60%	\$1,362,692	\$1,262,692
Regime-Aware Optimiser	\$100,000	10.38%	13.06%	0.66	-36.59%	\$1,843,697	\$1,743,697
Clean-Sheet Quant	\$100,000	8.97%	10.69%	0.66	-27.73%	\$1,260,577	\$1,160,577
DRL Raw Challenger	\$186,060,522	8.83%	10.58%	0.65	-27.87%	\$2,260,590,356	\$2,074,529,834
CVaR-Constrained Optimiser	\$100,000	9.72%	12.15%	0.65	-32.27%	\$1,541,859	\$1,441,859
Final Resolved Portfolio	\$186,060,522	9.09%	11.11%	0.65	-28.45%	\$2,424,094,460	\$2,238,033,938
DRL Baseline Portfolio	\$186,060,522	9.09%	11.11%	0.65	-28.45%	\$2,424,094,460	\$2,238,033,938
Dividend-Income Optimiser	\$100,000	10.51%	14.12%	0.63	-41.29%	\$1,906,226	\$1,806,226
Trend and Risk-Controlled Regional Indices	\$100,000	4.43%	8.61%	0.30	-23.30%	\$358,822	\$258,822

The performance table converts each assigned starting balance into a fully net ending balance. CAGR is the smooth annual growth rate, volatility measures return variation, Sharpe compares excess return with total variability, and maximum drawdown is the deepest peak-to-trough loss. Current Portfolio records the highest common-window CAGR at 14.20% alongside a -28.91% maximum drawdown.



Indexed growth uses each portfolio's assigned capital only for PnL; this chart rebases every series to 100 for comparability.



Paper Ratios and Annual Bank Charge

Performance Ratios on the Common Investable Window

Portfolio or index	cagr	sharpe	Lo Sharpe	sortino	calmar	Max drawdown	Monthly VaR 5%	Monthly ES 5%	PSR	MinTRL months	DSR
S&P 500 Index	12.52%	0.77	1.04	1.23	0.51	-24.77%	6.35%	8.77%			
Dow Jones Industrial Average	10.29%	0.63	0.91	1.01	0.44	-23.20%	5.42%	8.35%			
Current Portfolio	14.20%	0.80	0.85	1.32	0.49	-28.91%	6.99%	9.22%	99.99%	64.858	99.73%
LLM Analyst Benchmark	9.48%	0.76	0.79	1.37	0.62	-15.26%	3.69%	5.21%	100.00%	53.898	99.92%
Final Resolved Portfolio	9.25%	0.64	0.71	1.11	0.46	-19.91%	5.12%	6.70%	99.97%	83.862	98.92%
Clean-Sheet Quant	8.95%	0.64	0.62	1.14	0.43	-20.98%	4.51%	5.87%	99.98%	75.886	99.26%
DAX Performance Index	8.18%	0.42	0.45	0.65	0.22	-36.98%	8.02%	10.99%			

Portfolio or index	cagr	sharpe	Lo Sharpe	sortino	calmar	Max drawdown	Monthly VaR 5%	Monthly ES 5%	PSR	MinTRL months	DSR
Trend and Risk-Controlled Regional Indices	3.53%	0.23	0.22	0.33	0.15	-23.30%	4.19%	6.42%	93.87%	401.646	67.58%
FTSE 100 Index	3.06%	0.16	0.18	0.22	0.08	-38.76%	6.72%	9.76%			

Sharpe uses all volatility; Lo-adjusted Sharpe allows for serially related monthly returns; Sortino counts downside variability; and Calmar divides CAGR by maximum drawdown. Monthly VaR is the 5% loss threshold, while expected shortfall is the average loss beyond it. PSR measures confidence that Sharpe clears a hurdle, MinTRL estimates the required history, and DSR discounts non-normal returns and multiple trials. SPDR S&P 500 ETF adjusted-close proxy has the strongest Lo-adjusted Sharpe at 1.21.

Annual Fee Assumption

Charge	Annual rate	Basis points	Assessment month	Reference AUM	Bank charge	Calculated charge	Difference	Benchmarks charged
Annual bank custody and AUM charge	0.25%	25.000	12	\$186,060,522	\$465,151	\$465,151	\$0	False

The bank charge is 25 basis points, assessed once in month 12 on then-current AUM. At reference AUM of \$186,060,522, the configured and calculated charges equal \$465,151 and differ by \$0. The same percentage, not the full reference-dollar amount, applies to each portfolio; external benchmarks remain unchanged.

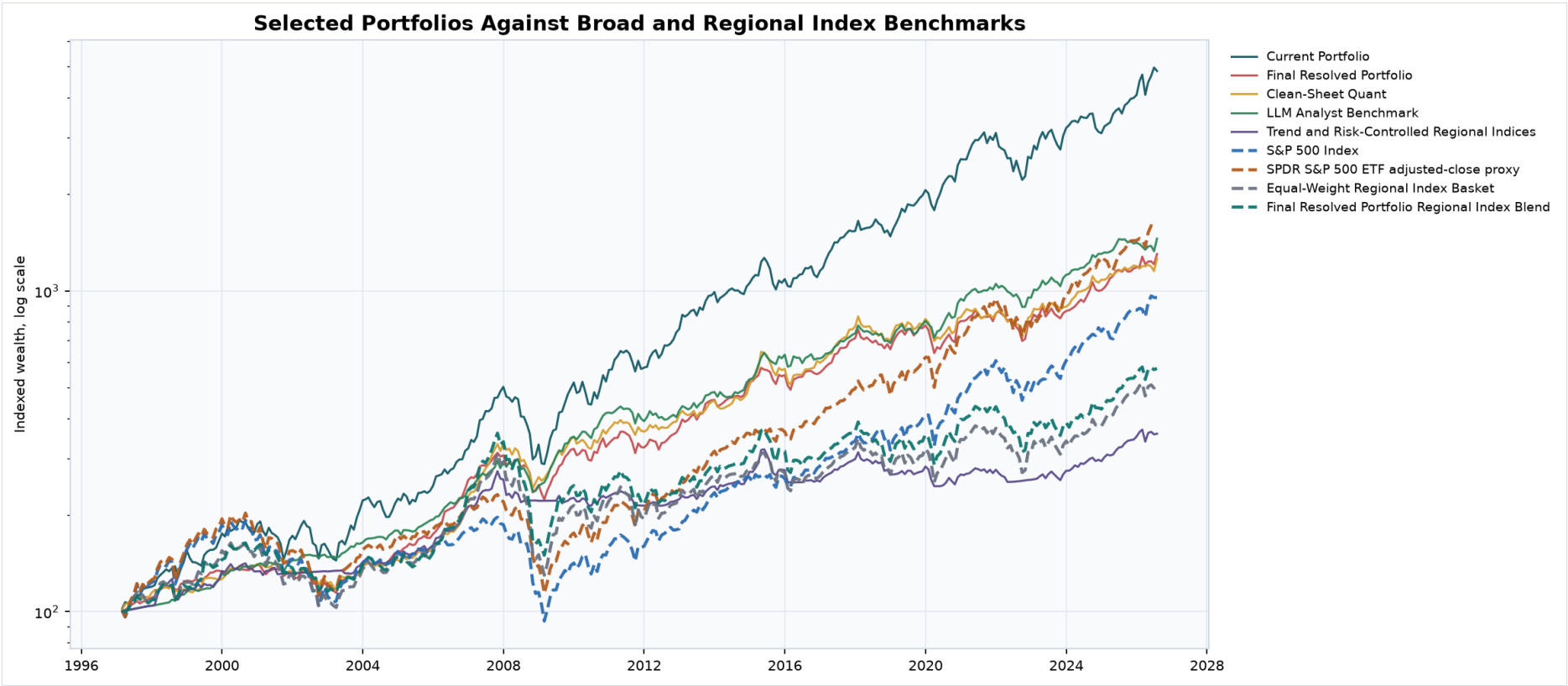
Fee Impact by Portfolio

Portfolio	Trading costs	Bank fees paid	Total modeled costs	Annual assessments	Ending value before fee	Ending value after fee	Ending-value fee drag	Effective fee bps
current_portfolio	\$33,220,087	\$155,185,066	\$188,405,153	29	\$9,691,705,955	\$9,013,105,522	\$678,600,433	25.000
portfolio_aware_overlay	\$46,663,105	\$120,844,335	\$167,507,440	29	\$7,280,532,797	\$6,770,759,519	\$509,773,278	25.000
final_portfolio	\$22,311,538	\$61,325,807	\$83,637,346	29	\$2,606,605,532	\$2,424,094,460	\$182,511,072	25.000

Portfolio	Trading costs	Bank fees paid	Total modeled costs	Annual assessments	Ending value before fee	Ending value after fee	Ending-value fee drag	Effective fee bps
drl_baseline	\$22,311,538	\$61,325,807	\$83,637,346	29	\$2,606,605,532	\$2,424,094,460	\$182,511,072	25.000
drl_challenger_raw	\$20,638,492	\$58,122,897	\$78,761,389	29	\$2,430,791,137	\$2,260,590,356	\$170,200,781	25.000
optimised_risk_parity	\$10,579	\$51,035	\$61,615	29	\$2,266,995	\$2,108,263	\$158,732	25.000
optimised_score_weighted	\$9,875	\$42,431	\$52,306	29	\$2,081,097	\$1,935,381	\$145,716	25.000
optimised_dividend_income	\$10,837	\$41,948	\$52,785	29	\$2,049,747	\$1,906,226	\$143,521	25.000
optimised_regime_aware	\$9,359	\$39,715	\$49,074	29	\$1,982,510	\$1,843,697	\$138,813	25.000
llm_benchmark	\$8,033	\$37,510	\$45,544	29	\$1,563,962	\$1,454,455	\$109,507	25.000
optimised_cvar_constrained	\$8,501	\$36,740	\$45,242	29	\$1,657,946	\$1,541,859	\$116,087	25.000
clean_sheet	\$5,784	\$34,161	\$39,945	29	\$1,355,486	\$1,260,577	\$94,909	25.000
optimised_mean_variance	\$7,155	\$33,532	\$40,686	29	\$1,465,289	\$1,362,692	\$102,598	25.000
trend_risk_controlled_indices	\$12,171	\$16,094	\$28,265	29	\$385,838	\$358,822	\$27,016	25.000

The fee-impact table separates trading costs from the annual bank charge and compares ending value before bank fees with the fully net result. current_portfolio pays the largest cumulative bank charge at \$155,185,066, with ending-value fee drag of \$678,600,433.

Major Index Benchmarks

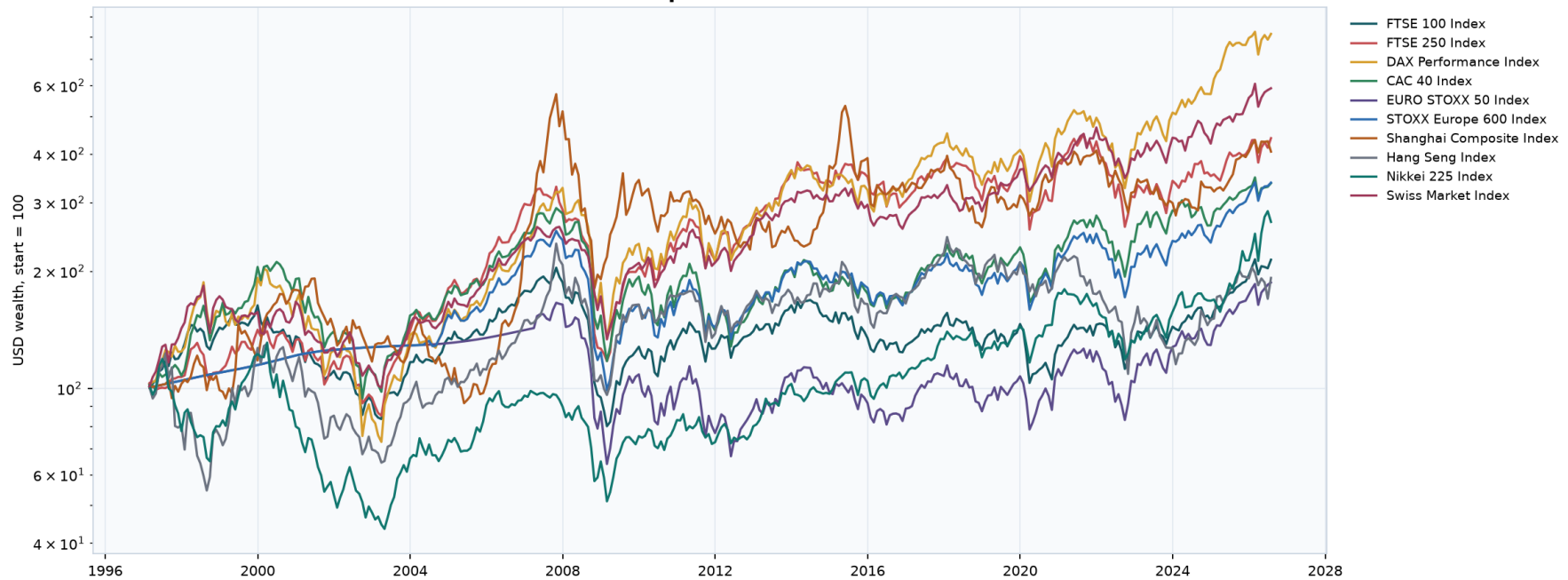


Dashed lines are broad or regional index benchmarks. Portfolio-specific relative statistics use each allocation's geographic index blend on the common investable window.

Major Index Benchmarks in USD United States



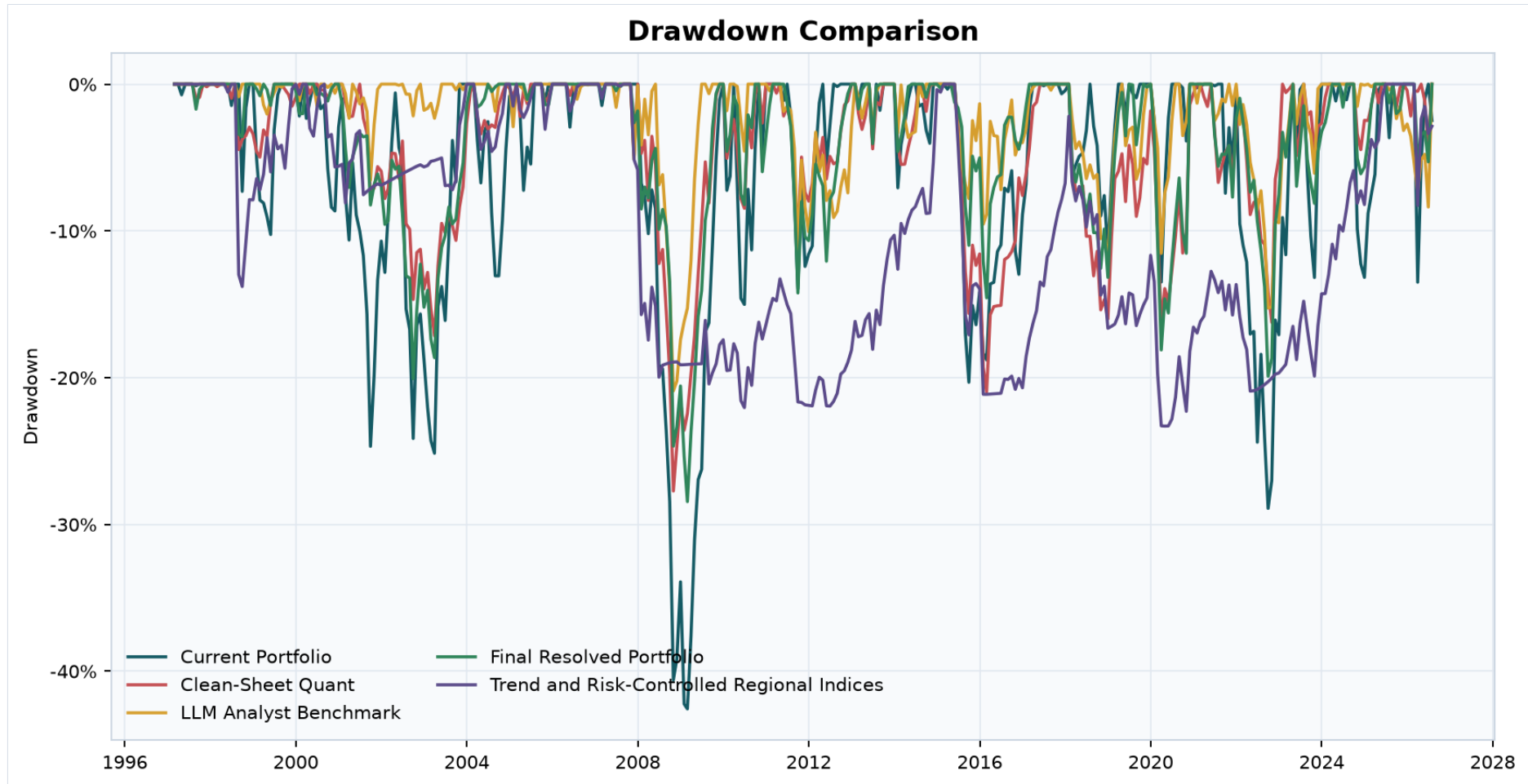
Europe and Asia

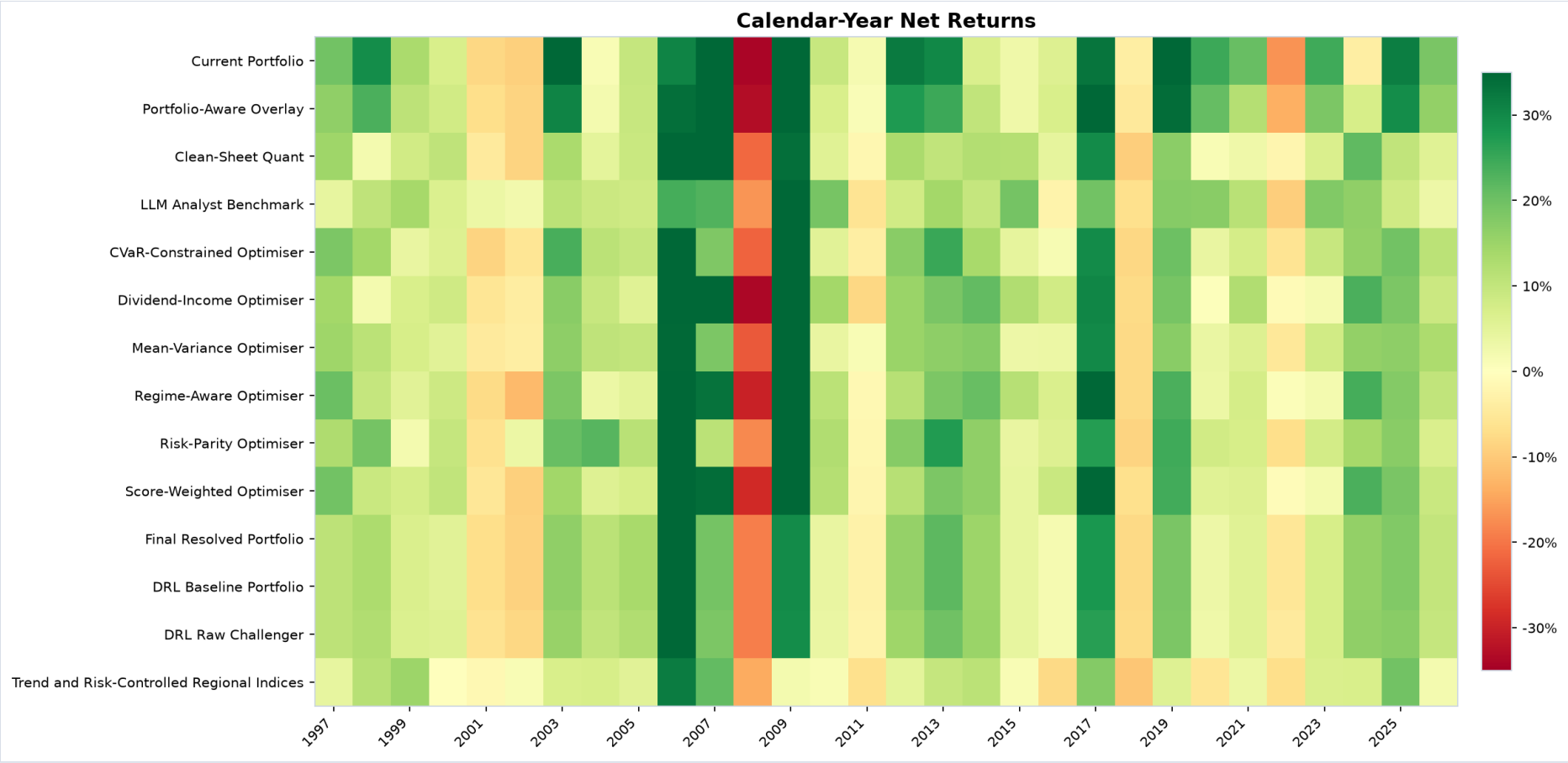


Standalone Index Performance, Common Investable Window

Index benchmark	cagr	Volatility	sharpe	Max drawdown	Ending value on \$100k
Nasdaq Composite Index	16.35%	17.66%	0.85	-33.10%	\$706,818
SPDR S&P 500 ETF adjusted-close proxy	14.41%	14.44%	0.88	-23.93%	\$569,250
S&P 500 Index	12.52%	14.45%	0.77	-24.77%	\$458,656
S&P 500 Index	12.52%	14.45%	0.77	-24.77%	\$458,656
iShares MSCI ACWI adjusted-close proxy	11.21%	13.97%	0.70	-25.74%	\$394,568
Dow Jones Industrial Average	10.29%	14.33%	0.63	-23.20%	\$354,382
Nikkei 225 Index	8.78%	17.09%	0.47	-34.01%	\$296,655
Russell 2000 Index	8.59%	19.63%	0.43	-33.76%	\$289,973
DAX Performance Index	8.18%	18.92%	0.42	-36.98%	\$276,092
iShares MSCI EAFE adjusted-close proxy	7.73%	14.50%	0.46	-27.62%	\$261,732
Equal-Weight Regional Index Basket	6.20%	14.44%	0.36	-28.51%	\$217,529
Swiss Market Index	6.05%	14.08%	0.36	-25.92%	\$213,444
EURO STOXX 50 Index	5.67%	18.40%	0.29	-34.44%	\$203,942
STOXX Europe 600 Index	5.12%	15.86%	0.28	-31.71%	\$190,599
CAC 40 Index	5.04%	17.91%	0.26	-30.37%	\$188,831
Shanghai Composite Index	3.98%	20.89%	0.20	-51.25%	\$165,543
FTSE 100 Index	3.06%	15.33%	0.16	-38.76%	\$147,582
FTSE 250 Index	2.79%	19.32%	0.15	-42.31%	\$142,755
Hang Seng Index	1.28%	20.58%	0.07	-55.51%	\$117,789

Each standalone index is converted to USD and replayed as an uncharged buy-and-hold reference on \$100,000. Price-index and total-return conventions differ, so these are broad path comparisons rather than perfectly dividend-consistent alpha tests. Nasdaq Composite Index has the highest common-window CAGR at 16.35%.





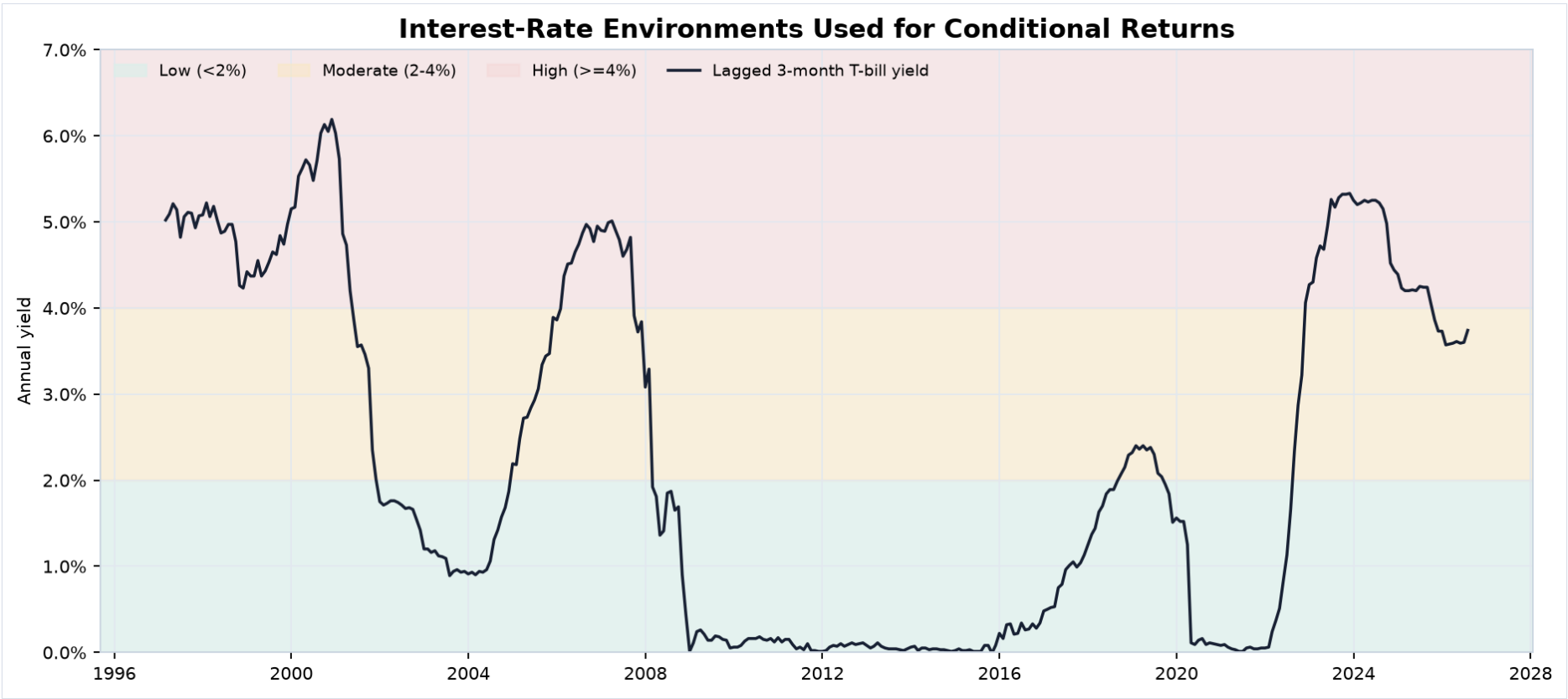
Portfolio-Relative Regional Benchmark Results

Portfolio	Regional benchmark	annualised_alpha	beta	tracking_error	information_ratio	active_return_cumulative	Relative PnL
Portfolio-Aware Overlay	Portfolio-Aware Overlay Regional Index Blend	8.35%	0.80	6.67%	1.04	152.50%	\$596,268,264
LLM Analyst Benchmark	LLM Analyst Benchmark Regional Index Blend	5.68%	0.87	5.88%	0.87	91.69%	\$154,097
Dividend-Income Optimiser	Dividend-Income Optimiser Regional Index Blend	6.98%	0.86	7.22%	0.84	120.95%	\$237,923

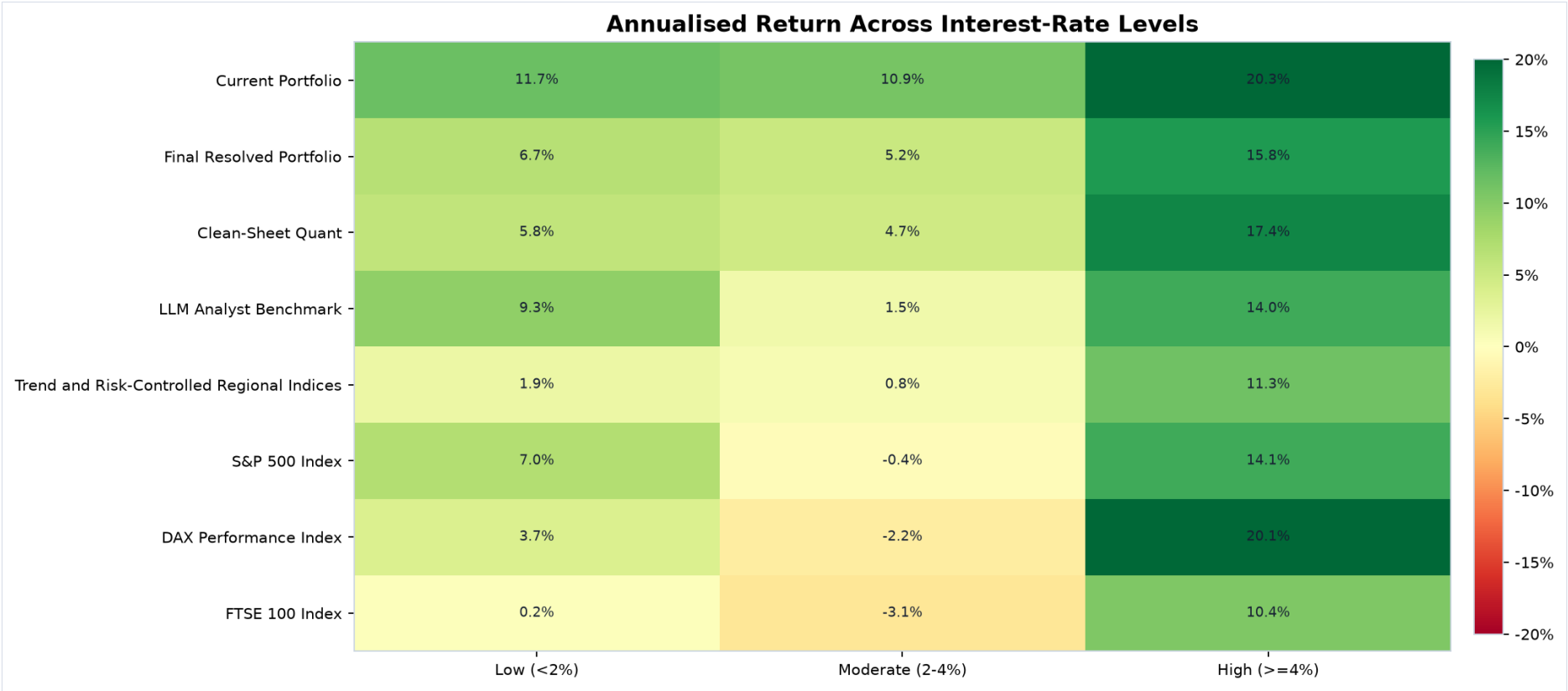
Portfolio	Regional benchmark	annualised_alpha	beta	tracking_error	information_ratio	active_return_cumulative	Relative PnL
Current Portfolio	Current Portfolio Regional Index Blend	8.63%	0.84	9.02%	0.83	161.81%	\$638,743,517
Regime-Aware Optimiser	Regime-Aware Optimiser Regional Index Blend	6.78%	0.82	7.63%	0.71	103.80%	\$233,225
Score-Weighted Optimiser	Score-Weighted Optimiser Regional Index Blend	6.53%	0.79	7.26%	0.70	98.21%	\$206,114
Clean-Sheet Quant	Clean-Sheet Quant Regional Index Blend	5.08%	0.79	7.29%	0.55	67.79%	\$122,217
CVaR-Constrained Optimiser	CVaR-Constrained Optimiser Regional Index Blend	4.68%	0.77	7.54%	0.40	51.26%	\$114,637
Mean-Variance Optimiser	Mean-Variance Optimiser Regional Index Blend	4.85%	0.72	7.46%	0.38	50.42%	\$110,461
DRL Raw Challenger	DRL Raw Challenger Regional Index Blend	4.40%	0.71	7.11%	0.33	41.23%	\$167,983,756
Final Resolved Portfolio	Final Resolved Portfolio Regional Index Blend	4.42%	0.71	7.54%	0.30	40.16%	\$167,082,596
DRL Baseline Portfolio	DRL Baseline Portfolio Regional Index Blend	4.42%	0.71	7.54%	0.30	40.16%	\$167,082,596
Risk-Parity Optimiser	Risk-Parity Optimiser Regional Index Blend	3.84%	0.74	6.92%	0.24	28.73%	\$75,412

Each portfolio is compared with an index blend matching its regional weights. Alpha is annualized return not explained by beta, tracking error is active-return volatility, and information ratio is active return per unit of tracking error. Portfolio-Aware Overlay has the highest information ratio at 1.04 and relative PnL of \$596,268,264.

Interest Rates and Market Regimes



Interest-Rate Level Performance



Environment	Portfolio or index	Months	Conditional annual return	Volatility	sharpe	Conditional max drawdown	Worst month
High (>=4%)	Clean-Sheet Quant	105	17.37%	8.91%	1.31	-4.98%	-0.049
High (>=4%)	Current Portfolio	105	20.29%	14.14%	1.05	-13.18%	-0.099
High (>=4%)	DAX Performance Index	105	20.05%	19.16%	0.81	-30.45%	-0.157
High (>=4%)	Dow Jones Industrial Average	105	12.24%	14.69%	0.54	-16.82%	-0.151
High (>=4%)	FTSE 100 Index	105	10.39%	13.22%	0.45	-28.49%	-0.093
High (>=4%)	Final Resolved Portfolio	105	15.77%	8.87%	1.16	-8.13%	-0.070

Environment	Portfolio or index	Months	Conditional annual return	Volatility	sharpe	Conditional max drawdown	Worst month
High (>=4%)	LLM Analyst Benchmark	105	14.05%	6.36%	1.35	-6.08%	-0.027
High (>=4%)	S&P 500 Index	105	14.06%	14.87%	0.64	-23.55%	-0.146
High (>=4%)	Trend and Risk-Controlled Regional Indices	105	11.26%	8.54%	0.74	-13.80%	-0.130
Low (<2%)	Clean-Sheet Quant	198	5.84%	11.46%	0.50	-25.07%	-0.100
Low (<2%)	Current Portfolio	198	11.69%	17.62%	0.68	-41.16%	-0.170
Low (<2%)	DAX Performance Index	198	3.66%	24.69%	0.25	-55.93%	-0.249
Low (<2%)	Dow Jones Industrial Average	198	6.21%	15.05%	0.44	-44.99%	-0.141
Low (<2%)	FTSE 100 Index	198	0.16%	17.85%	0.07	-54.74%	-0.189
Low (<2%)	Final Resolved Portfolio	198	6.71%	12.20%	0.55	-25.17%	-0.129
Low (<2%)	LLM Analyst Benchmark	198	9.27%	9.82%	0.89	-20.91%	-0.083
Low (<2%)	S&P 500 Index	198	7.01%	15.65%	0.48	-47.51%	-0.169
Low (<2%)	Trend and Risk-Controlled Regional Indices	198	1.89%	7.78%	0.21	-21.14%	-0.083
Moderate (2-4%)	Clean-Sheet Quant	51	4.74%	10.52%	0.21	-11.89%	-0.049
Moderate (2-4%)	Current Portfolio	51	10.89%	17.74%	0.50	-20.42%	-0.135
Moderate (2-4%)	DAX Performance Index	51	-2.23%	21.11%	-0.14	-32.48%	-0.169
Moderate (2-4%)	Dow Jones Industrial Average	51	0.88%	15.93%	-0.06	-20.77%	-0.111
Moderate (2-4%)	FTSE 100 Index	51	-3.06%	15.01%	-0.33	-35.31%	-0.095
Moderate (2-4%)	Final Resolved Portfolio	51	5.18%	10.63%	0.25	-13.43%	-0.071
Moderate (2-4%)	LLM Analyst Benchmark	51	1.52%	10.44%	-0.09	-19.74%	-0.057
Moderate (2-4%)	S&P 500 Index	51	-0.36%	15.35%	-0.14	-23.13%	-0.093
Moderate (2-4%)	Trend and Risk-Controlled Regional Indices	51	0.82%	11.10%	-0.14	-22.32%	-0.105

This table groups non-contiguous months by the lagged 3-month Treasury-bill yield level and annualizes the observations in each group. It is a conditional description, not a continuous investable path or a causal estimate. For Final Resolved Portfolio, the strongest group is High ($\geq 4\%$), with a conditional geometric return of 15.77% across 105 months.

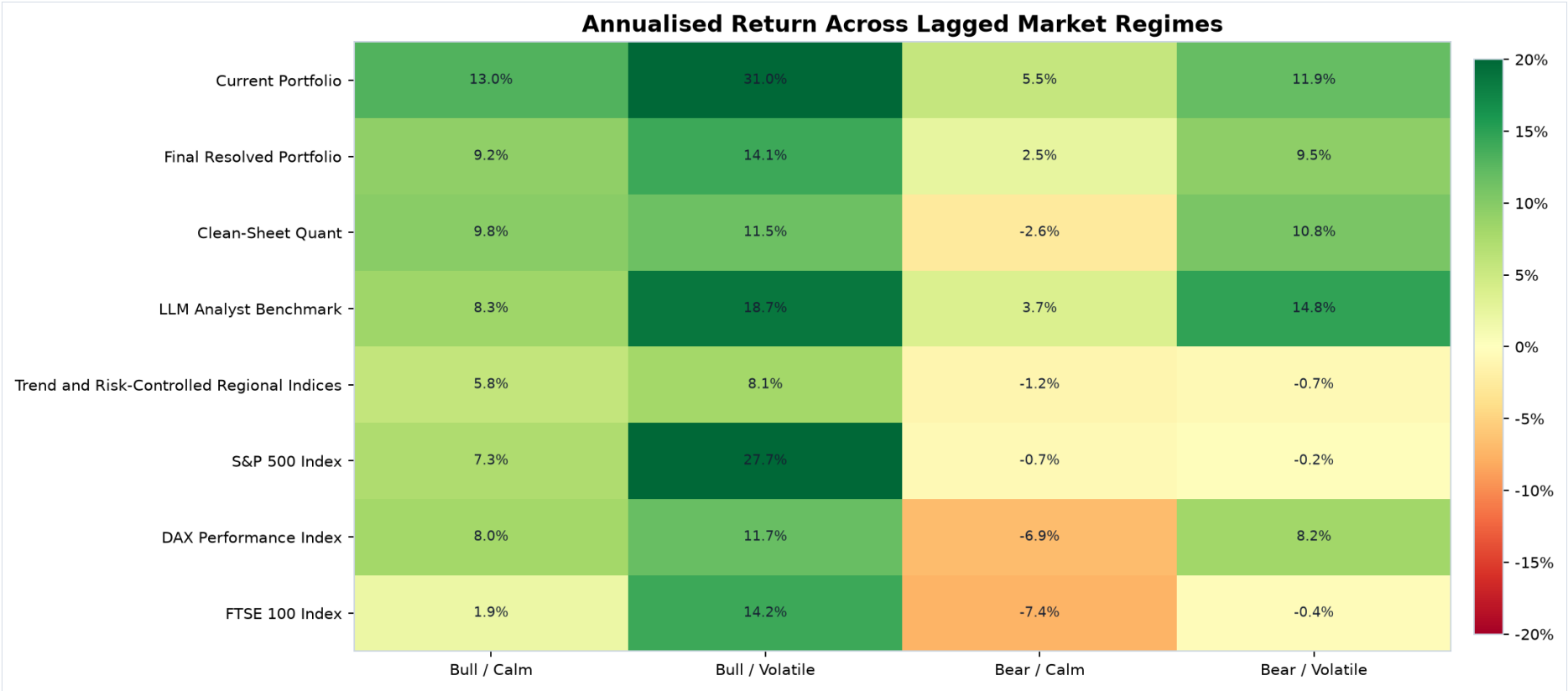
Interest-Rate Direction Performance

Environment	Portfolio or index	Months	Conditional annual return	Volatility	sharpe	Conditional max drawdown	Worst month
Falling	Clean-Sheet Quant	78	2.06%	12.37%	0.08	-28.79%	-0.100
Falling	Current Portfolio	78	7.61%	20.88%	0.37	-42.57%	-0.170
Falling	DAX Performance Index	78	-3.92%	30.32%	-0.04	-73.09%	-0.249
Falling	Dow Jones Industrial Average	78	-2.22%	20.19%	-0.10	-58.19%	-0.141
Falling	FTSE 100 Index	78	-8.48%	21.39%	-0.39	-68.40%	-0.189
Falling	Final Resolved Portfolio	78	1.67%	13.99%	0.05	-32.42%	-0.129
Falling	LLM Analyst Benchmark	78	10.25%	11.23%	0.76	-20.91%	-0.081
Falling	S&P 500 Index	78	-2.40%	20.26%	-0.11	-64.25%	-0.169
Falling	Trend and Risk-Controlled Regional Indices	78	1.56%	9.38%	0.01	-26.93%	-0.105
Rising	Clean-Sheet Quant	79	13.47%	10.52%	0.91	-16.01%	-0.054
Rising	Current Portfolio	79	15.36%	15.44%	0.77	-14.51%	-0.091
Rising	DAX Performance Index	79	7.38%	18.26%	0.28	-37.01%	-0.133
Rising	Dow Jones Industrial Average	79	7.25%	14.15%	0.30	-14.64%	-0.088
Rising	FTSE 100 Index	79	2.23%	14.61%	-0.03	-28.15%	-0.095
Rising	Final Resolved Portfolio	79	13.03%	10.65%	0.86	-15.22%	-0.071
Rising	LLM Analyst Benchmark	79	9.61%	8.32%	0.71	-11.93%	-0.057
Rising	S&P 500 Index	79	6.46%	14.20%	0.25	-15.59%	-0.093

Environment	Portfolio or index	Months	Conditional annual return	Volatility	sharpe	Conditional max drawdown	Worst month
Rising	Trend and Risk-Controlled Regional Indices	79	6.40%	7.62%	0.37	-14.78%	-0.060
Stable	Clean-Sheet Quant	185	9.59%	10.23%	0.81	-20.98%	-0.087
Stable	Current Portfolio	185	15.88%	15.67%	0.93	-20.32%	-0.135
Stable	DAX Performance Index	185	11.27%	21.18%	0.54	-33.56%	-0.191
Stable	Dow Jones Industrial Average	185	10.78%	12.60%	0.76	-16.82%	-0.151
Stable	FTSE 100 Index	185	6.20%	14.35%	0.39	-26.60%	-0.120
Stable	Final Resolved Portfolio	185	10.54%	10.20%	0.89	-14.56%	-0.074
Stable	LLM Analyst Benchmark	185	9.45%	8.69%	0.92	-11.53%	-0.083
Stable	S&P 500 Index	185	12.23%	13.32%	0.83	-17.03%	-0.146
Stable	Trend and Risk-Controlled Regional Indices	185	4.79%	8.97%	0.41	-21.14%	-0.130

This table groups non-contiguous months by the prior 12-month change in the Treasury-bill yield and annualizes the observations in each group. It is a conditional description, not a continuous investable path or a causal estimate. For Final Resolved Portfolio, the strongest group is Rising, with a conditional geometric return of 13.03% across 79 months.

Lagged Market Regime Performance



Environment	Portfolio or index	Months	Conditional annual return	Volatility	sharpe	Conditional max drawdown	Worst month
Bear / Calm	Clean-Sheet Quant	37	-2.56%	12.74%	-0.25	-21.12%	-0.087
Bear / Calm	Current Portfolio	37	5.52%	19.92%	0.29	-31.26%	-0.117
Bear / Calm	DAX Performance Index	37	-6.86%	30.49%	-0.13	-43.42%	-0.249
Bear / Calm	Dow Jones Industrial Average	37	2.66%	19.16%	0.15	-28.86%	-0.124
Bear / Calm	FTSE 100 Index	37	-7.36%	20.01%	-0.36	-37.14%	-0.149
Bear / Calm	Final Resolved Portfolio	37	2.48%	13.39%	0.14	-15.71%	-0.079

Environment	Portfolio or index	Months	Conditional annual return	Volatility	sharpe	Conditional max drawdown	Worst month
Bear / Calm	LLM Analyst Benchmark	37	3.71%	11.60%	0.24	-13.97%	-0.083
Bear / Calm	S&P 500 Index	37	-0.66%	19.82%	-0.01	-33.63%	-0.110
Bear / Calm	Trend and Risk-Controlled Regional Indices	37	-1.15%	7.38%	-0.32	-10.36%	-0.083
Bear / Volatile	Clean-Sheet Quant	43	10.76%	12.91%	0.71	-16.48%	-0.100
Bear / Volatile	Current Portfolio	43	11.95%	23.90%	0.51	-32.92%	-0.170
Bear / Volatile	DAX Performance Index	43	8.22%	34.18%	0.35	-54.12%	-0.230
Bear / Volatile	Dow Jones Industrial Average	43	2.21%	21.32%	0.12	-41.77%	-0.141
Bear / Volatile	FTSE 100 Index	43	-0.45%	22.73%	0.01	-48.85%	-0.189
Bear / Volatile	Final Resolved Portfolio	43	9.49%	15.11%	0.55	-19.55%	-0.129
Bear / Volatile	LLM Analyst Benchmark	43	14.75%	10.93%	1.14	-8.63%	-0.081
Bear / Volatile	S&P 500 Index	43	-0.22%	22.29%	0.02	-48.15%	-0.169
Bear / Volatile	Trend and Risk-Controlled Regional Indices	43	-0.68%	4.82%	-0.50	-7.60%	-0.052
Bull / Calm	Clean-Sheet Quant	226	9.82%	10.14%	0.76	-29.36%	-0.070
Bull / Calm	Current Portfolio	226	13.05%	14.73%	0.76	-26.80%	-0.135
Bull / Calm	DAX Performance Index	226	8.05%	18.98%	0.39	-49.34%	-0.191
Bull / Calm	Dow Jones Industrial Average	226	5.86%	12.95%	0.33	-27.35%	-0.151
Bull / Calm	FTSE 100 Index	226	1.88%	14.35%	0.05	-55.38%	-0.160
Bull / Calm	Final Resolved Portfolio	226	9.23%	10.22%	0.70	-26.68%	-0.082
Bull / Calm	LLM Analyst Benchmark	226	8.34%	8.18%	0.75	-17.36%	-0.075
Bull / Calm	S&P 500 Index	226	7.30%	12.96%	0.44	-27.31%	-0.146
Bull / Calm	Trend and Risk-Controlled Regional Indices	226	5.78%	9.78%	0.40	-27.71%	-0.130
Bull / Volatile	Clean-Sheet Quant	36	11.52%	10.21%	0.92	-4.51%	-0.045

Environment	Portfolio or index	Months	Conditional annual return	Volatility	sharpe	Conditional max drawdown	Worst month
Bull / Volatile	Current Portfolio	36	30.99%	16.64%	1.59	-7.91%	-0.073
Bull / Volatile	DAX Performance Index	36	11.74%	21.28%	0.53	-17.16%	-0.103
Bull / Volatile	Dow Jones Industrial Average	36	24.63%	13.06%	1.60	-6.78%	-0.067
Bull / Volatile	FTSE 100 Index	36	14.19%	14.68%	0.84	-9.34%	-0.065
Bull / Volatile	Final Resolved Portfolio	36	14.09%	10.30%	1.14	-6.35%	-0.060
Bull / Volatile	LLM Analyst Benchmark	36	18.74%	10.29%	1.53	-5.16%	-0.042
Bull / Volatile	S&P 500 Index	36	27.74%	13.87%	1.70	-8.26%	-0.066
Bull / Volatile	Trend and Risk-Controlled Regional Indices	36	8.10%	6.43%	0.93	-4.57%	-0.030

This table groups non-contiguous months by lagged S&P 500 momentum and volatility and annualizes the observations in each group. It is a conditional description, not a continuous investable path or a causal estimate. For Final Resolved Portfolio, the strongest group is Bull / Volatile, with a conditional geometric return of 14.09% across 36 months.

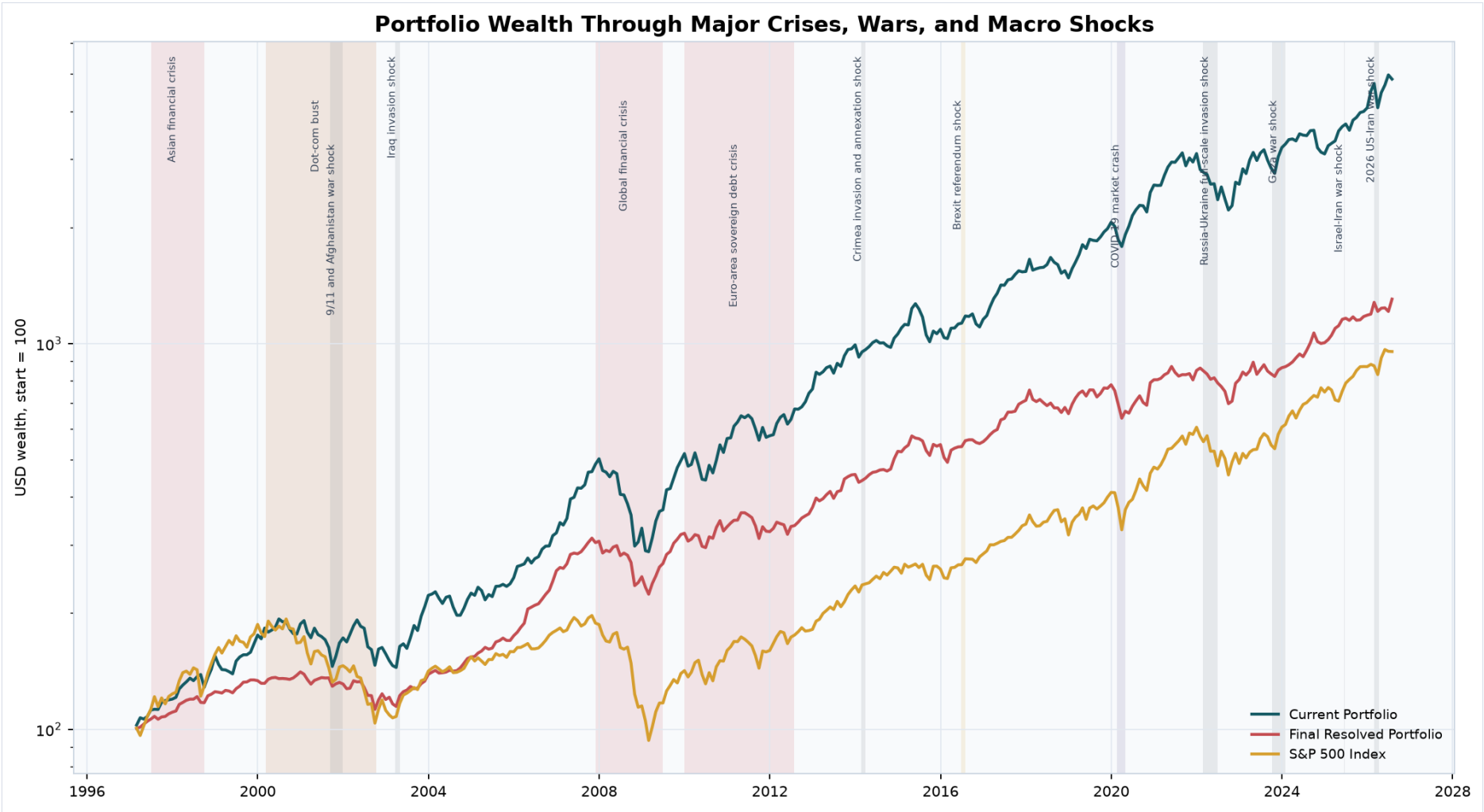
Retrospective Economic-Cycle Performance

Environment	Portfolio or index	Months	Conditional annual return	Volatility	sharpe	Conditional max drawdown	Worst month
Expansion	Clean-Sheet Quant	326	9.86%	10.40%	0.75	-20.98%	-0.087
Expansion	Current Portfolio	326	15.57%	15.66%	0.86	-28.91%	-0.135
Expansion	DAX Performance Index	326	9.33%	21.25%	0.42	-56.23%	-0.249
Expansion	Dow Jones Industrial Average	326	8.78%	14.15%	0.51	-29.27%	-0.151
Expansion	FTSE 100 Index	326	4.66%	15.18%	0.23	-42.58%	-0.160
Expansion	Final Resolved Portfolio	326	9.91%	10.76%	0.73	-19.91%	-0.082
Expansion	LLM Analyst Benchmark	326	9.59%	8.71%	0.84	-15.26%	-0.083

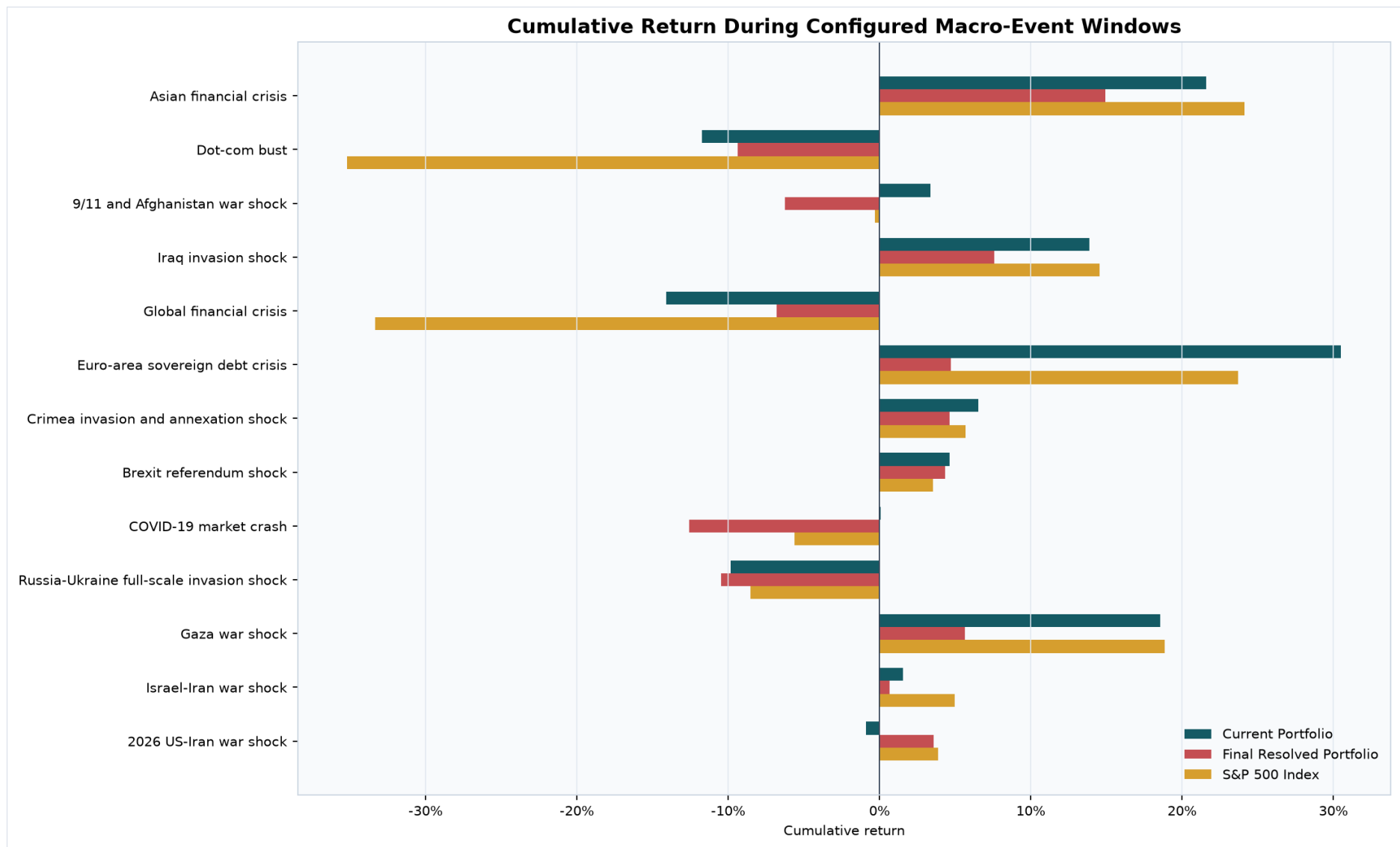
Environment	Portfolio or index	Months	Conditional annual return	Volatility	sharpe	Conditional max drawdown	Worst month
Expansion	S&P 500 Index	326	9.67%	14.35%	0.56	-41.54%	-0.146
Expansion	Trend and Risk-Controlled Regional Indices	326	4.91%	8.75%	0.34	-23.29%	-0.130
Recession	Clean-Sheet Quant	28	-0.86%	13.59%	-0.09	-28.65%	-0.100
Recession	Current Portfolio	28	-2.18%	25.69%	-0.01	-41.89%	-0.170
Recession	DAX Performance Index	28	-13.09%	35.82%	-0.25	-60.81%	-0.230
Recession	Dow Jones Industrial Average	28	-9.91%	23.11%	-0.39	-48.72%	-0.141
Recession	FTSE 100 Index	28	-18.51%	24.96%	-0.74	-58.45%	-0.189
Recession	Final Resolved Portfolio	28	0.02%	14.63%	-0.02	-25.17%	-0.129
Recession	LLM Analyst Benchmark	28	8.45%	12.78%	0.59	-20.91%	-0.081
Recession	S&P 500 Index	28	-10.34%	24.24%	-0.38	-51.25%	-0.169
Recession	Trend and Risk-Controlled Regional Indices	28	-1.02%	6.72%	-0.31	-8.70%	-0.057

This table groups non-contiguous months by the retrospective NBER recession indicator and annualizes the observations in each group. It is a conditional description, not a continuous investable path or a causal estimate. For Final Resolved Portfolio, the strongest group is Expansion, with a conditional geometric return of 9.91% across 326 months.

Major Macro Events



Shading includes the Asian crisis, dot-com bust, global financial crisis, COVID-19, political shocks, and war windows through the 2026 US-Iran conflict.



Configured Event Windows

Event window	Type	Start	End	Days
Asian financial crisis	financial_crisis	1997-07-02 00:00:00	1998-09-30 00:00:00	456

Event window	Type	Start	End	Days
Dot-com bust	market_crash	2000-03-10 00:00:00	2002-10-09 00:00:00	944
9/11 and Afghanistan war shock	war	2001-09-11 00:00:00	2001-12-31 00:00:00	112
Iraq invasion shock	war	2003-03-20 00:00:00	2003-05-01 00:00:00	43
Global financial crisis	financial_crisis	2007-12-01 00:00:00	2009-06-30 00:00:00	578
Euro-area sovereign debt crisis	financial_crisis	2010-01-01 00:00:00	2012-07-26 00:00:00	938
Crimea invasion and annexation shock	war	2014-02-20 00:00:00	2014-03-31 00:00:00	40
Brexit referendum shock	political_shock	2016-06-23 00:00:00	2016-07-31 00:00:00	39
COVID-19 market crash	pandemic	2020-02-19 00:00:00	2020-04-30 00:00:00	72
Russia-Ukraine full-scale invasion shock	war	2022-02-24 00:00:00	2022-06-30 00:00:00	127
Gaza war shock	war	2023-10-07 00:00:00	2024-01-31 00:00:00	117
Israel-Iran war shock	war	2025-06-13 00:00:00	2025-06-24 00:00:00	12
2026 US-Iran war shock	war	2026-02-28 00:00:00	2026-04-08 00:00:00	40

The event-definition table supplies source-backed windows shaded on the charts. They are market-response intervals for monthly analysis, not claims about legal war dates, and overlapping events can share return months.

Portfolio and S&P 500 Event Performance

Event window	Portfolio or index	Months	cumulative_return	Max drawdown	worst_month	Starting AUM	Event PnL
Asian financial crisis	Current Portfolio	16	21.58%	-7.31%	-0.073	\$208,623,099	\$45,021,576
Asian financial crisis	Final Resolved Portfolio	16	14.92%	-3.58%	-0.034	\$197,648,402	\$29,487,886
Asian financial crisis	S&P 500 Index	16	24.12%	-15.57%	-0.146	\$112,590	\$27,161
Dot-com bust	Current Portfolio	32	-11.74%	-24.68%	-0.107	\$340,136,051	\$-39,939,229

Event window	Portfolio or index	Months	cumulative_return	Max drawdown	worst_month	Starting AUM	Event PnL
Dot-com bust	Final Resolved Portfolio	32	-9.36%	-20.07%	-0.079	\$244,648,873	\$-22,905,405
Dot-com bust	S&P 500 Index	32	-35.18%	-46.28%	-0.110	\$173,809	\$-61,140
9/11 and Afghanistan war shock	Current Portfolio	5	3.37%	-10.69%	-0.107	\$302,973,491	\$10,203,066
9/11 and Afghanistan war shock	Final Resolved Portfolio	5	-6.26%	-6.26%	-0.049	\$252,729,972	\$-15,819,534
9/11 and Afghanistan war shock	S&P 500 Index	5	-0.30%	-8.17%	-0.082	\$144,192	\$-430
Iraq invasion shock	Current Portfolio	3	13.87%	-1.14%	-0.011	\$271,993,587	\$37,735,080
Iraq invasion shock	Final Resolved Portfolio	3	7.57%	-1.53%	-0.015	\$216,415,806	\$16,381,426
Iraq invasion shock	S&P 500 Index	3	14.56%	0.00%	0.008	\$106,995	\$15,574
Global financial crisis	Current Portfolio	20	-14.12%	-42.57%	-0.170	\$904,465,314	\$-127,694,095
Global financial crisis	Final Resolved Portfolio	20	-6.82%	-27.12%	-0.129	\$566,663,906	\$-38,648,323
Global financial crisis	S&P 500 Index	20	-33.33%	-50.37%	-0.169	\$188,402	\$-62,794
Euro-area sovereign debt crisis	Current Portfolio	31	30.51%	-15.02%	-0.082	\$964,074,628	\$294,138,510
Euro-area sovereign debt crisis	Final Resolved Portfolio	31	4.70%	-14.23%	-0.072	\$599,191,736	\$28,188,959
Euro-area sovereign debt crisis	S&P 500 Index	31	23.69%	-17.03%	-0.082	\$141,841	\$33,609
Crimea invasion and annexation shock	Current Portfolio	3	6.52%	0.00%	0.014	\$1,714,710,132	\$111,832,596
Crimea invasion and annexation shock	Final Resolved Portfolio	3	4.65%	0.00%	0.011	\$811,980,936	\$37,762,085
Crimea invasion and annexation shock	S&P 500 Index	3	5.69%	0.00%	0.006	\$226,746	\$12,893
Brexit referendum shock	Current Portfolio	3	4.61%	-0.26%	-0.003	\$2,089,451,173	\$96,398,017
Brexit referendum shock	Final Resolved Portfolio	3	4.35%	0.00%	0.002	\$1,003,361,493	\$43,668,171
Brexit referendum shock	S&P 500 Index	3	3.53%	-0.12%	-0.001	\$266,733	\$9,413
COVID-19 market crash	Current Portfolio	4	0.09%	-11.36%	-0.076	\$3,743,959,481	\$3,486,474
COVID-19 market crash	Final Resolved Portfolio	4	-12.58%	-15.19%	-0.082	\$1,403,827,789	\$-176,631,304

Event window	Portfolio or index	Months	cumulative_return	Max drawdown	worst_month	Starting AUM	Event PnL
COVID-19 market crash	S&P 500 Index	4	-5.62%	-19.87%	-0.125	\$410,288	\$-23,050
Russia-Ukraine full-scale invasion shock	Current Portfolio	6	-9.82%	-16.45%	-0.091	\$5,251,174,747	\$-515,651,046
Russia-Ukraine full-scale invasion shock	Final Resolved Portfolio	6	-10.46%	-10.46%	-0.031	\$1,606,059,491	\$-168,020,124
Russia-Ukraine full-scale invasion shock	S&P 500 Index	6	-8.53%	-16.45%	-0.088	\$574,381	\$-49,005
Gaza war shock	Current Portfolio	5	18.54%	-3.13%	-0.031	\$5,299,623,310	\$982,632,625
Gaza war shock	Final Resolved Portfolio	5	5.62%	-1.50%	-0.015	\$1,551,324,158	\$87,227,261
Gaza war shock	S&P 500 Index	5	18.85%	-2.20%	-0.022	\$545,442	\$102,806
Israel-Iran war shock	Current Portfolio	1	1.54%	0.00%	0.015	\$6,789,721,030	\$104,830,843
Israel-Iran war shock	Final Resolved Portfolio	1	0.66%	0.00%	0.007	\$2,148,554,848	\$14,098,451
Israel-Iran war shock	S&P 500 Index	1	4.96%	0.00%	0.050	\$751,970	\$37,303
2026 US-Iran war shock	Current Portfolio	3	-0.92%	-13.50%	-0.135	\$8,391,562,322	\$-76,807,608
2026 US-Iran war shock	Final Resolved Portfolio	3	3.57%	-5.27%	-0.053	\$2,216,743,661	\$79,203,167
2026 US-Iran war shock	S&P 500 Index	3	3.89%	-5.92%	-0.051	\$882,649	\$34,342

Event return compounds all overlapping monthly observations, event drawdown measures the loss within that window, and event PnL applies the return to actual simulated AUM at the start of the window. The weakest configured event for the final portfolio is COVID-19 market crash, returning -12.58% with a -15.19% within-window drawdown.

The Three Backtests

1. Historical / walk-forward

Monthly execution with one-period signal lag for the index challenger.
Current portfolio outputs are shown only as retrospective holdings replays
and are explicitly flagged for selection look-ahead and survivorship bias.

2. Resampling

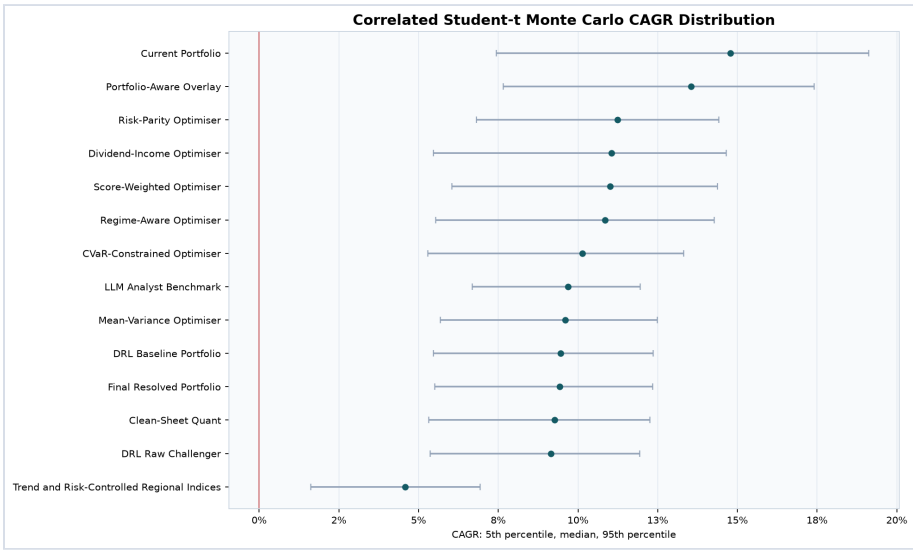
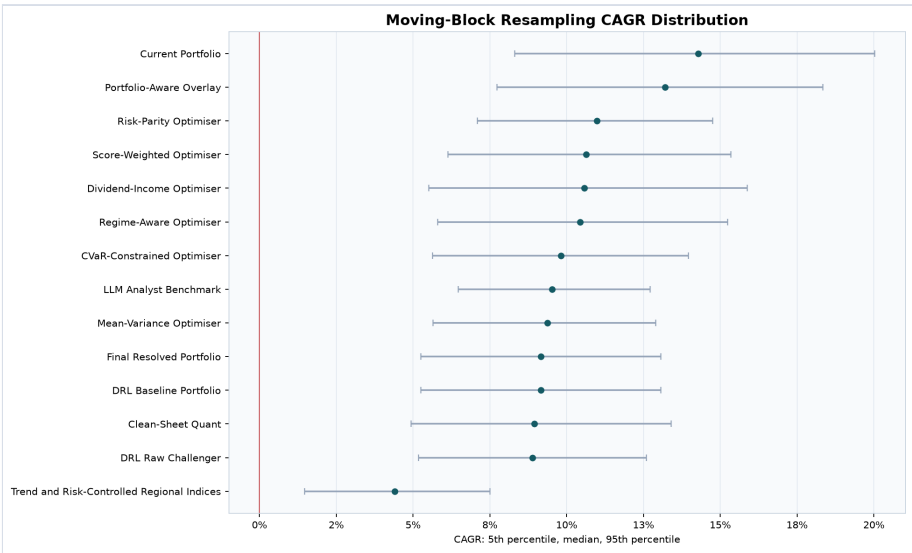
Circular 12-month moving blocks preserve serial dependence and cross-strategy alignment. The intervals show how path order changes CAGR, Sharpe, drawdown, and benchmark outperformance.

3. Monte Carlo

A correlated Student-t AR(1) process with EWMA conditional variance preserves cross-portfolio dependence, autocorrelation, fat tails, and volatility clustering without fitting a return target.

Statistical integrity

PSR, MinTRL, Sidak family-wise control, and Deflated Sharpe Ratios account for sample length, non-normal returns, and the number and correlation of portfolio trials.



Moving-Block Resampling Summary

Portfolio	cagr_p05	cagr_median	cagr_p95	Drawdown 5th percentile	Loss probability	Regional outperformance
current_portfolio	8.31%	14.27%	20.02%	-57.20%	0.00%	100.00%
portfolio_aware_overlay	7.73%	13.20%	18.34%	-53.64%	0.00%	100.00%
optimised_risk_parity	7.10%	10.99%	14.75%	-40.17%	0.00%	95.76%
optimised_score_weighted	6.13%	10.63%	15.35%	-46.98%	0.00%	99.30%
optimised_dividend_income	5.51%	10.58%	15.88%	-53.95%	0.02%	99.50%
optimised_regime_aware	5.80%	10.43%	15.24%	-48.53%	0.00%	98.42%
optimised_cvar_constrained	5.64%	9.81%	13.96%	-43.79%	0.00%	97.32%
llm_benchmark	6.47%	9.52%	12.72%	-29.00%	0.00%	99.94%
optimised_mean_variance	5.65%	9.37%	12.90%	-40.23%	0.00%	93.76%
final_portfolio	5.25%	9.16%	13.07%	-39.92%	0.00%	93.94%
drl_baseline	5.25%	9.16%	13.07%	-39.92%	0.00%	93.94%
clean_sheet	4.93%	8.95%	13.39%	-39.85%	0.00%	99.00%
drl_challenger_raw	5.17%	8.88%	12.59%	-38.52%	0.00%	94.30%
trend_risk_controlled_indices	1.46%	4.40%	7.50%	-40.49%	0.64%	27.04%

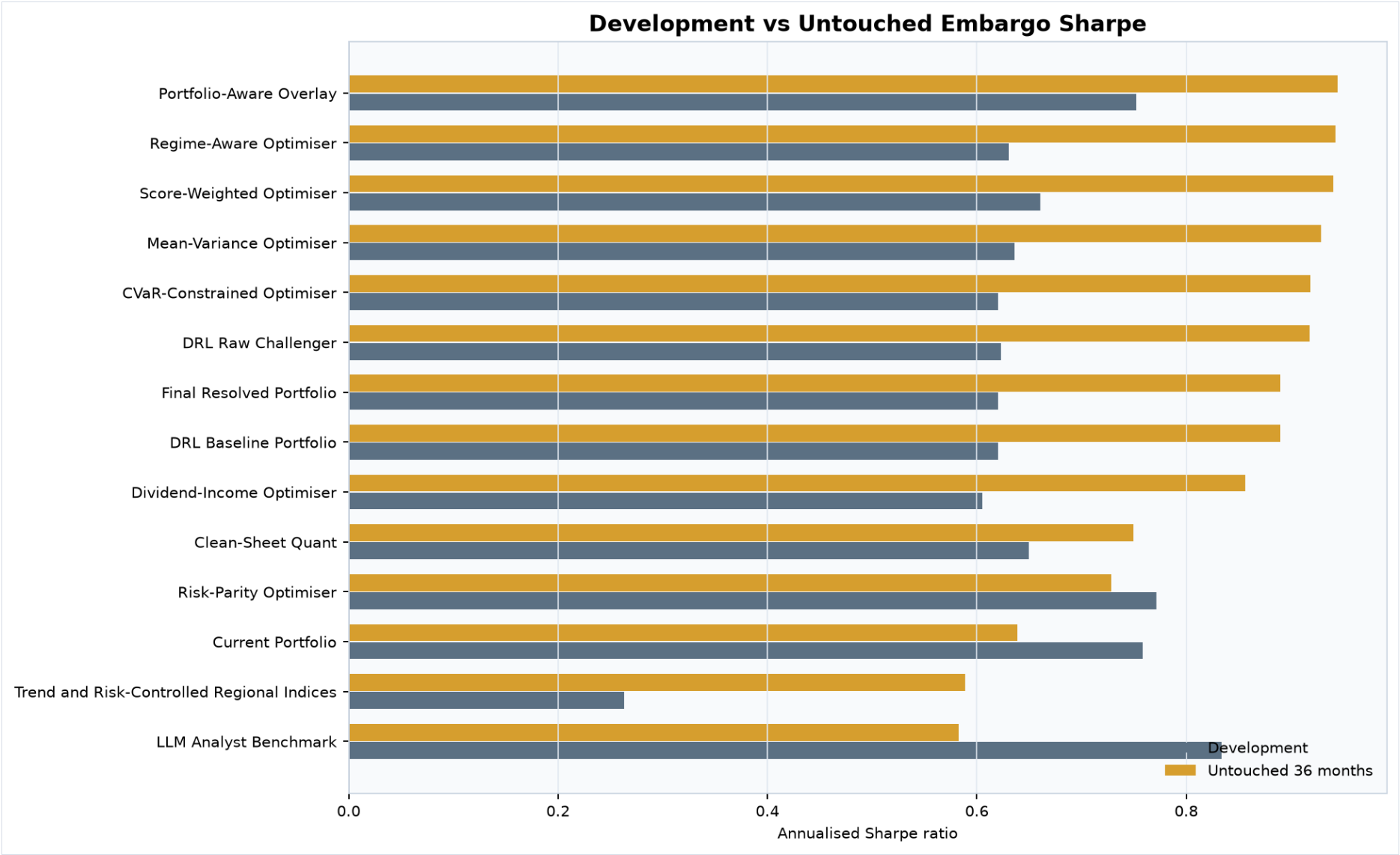
The moving-block bootstrap rearranges 12-month blocks while preserving short-run dependence and cross-strategy alignment. Its 5th, median, and 95th percentiles show sensitivity to historical path order. The lowest 5th-percentile CAGR is trend_risk_controlled_indices at 1.46%.

Monte Carlo Summary

Portfolio	cagr_p05	cagr_median	cagr_p95	Drawdown 5th percentile	Ending value 5th percentile	Median ending value	Ending value 95th percentile	Loss probability
current_portfolio	7.44%	14.78%	19.12%	-75.79%	\$1,543,608,689	\$10,862,738,865	\$32,433,946,766	1.06%
portfolio_aware_overlay	7.66%	13.54%	17.40%	-68.88%	\$1,639,972,773	\$7,888,900,317	\$21,152,170,060	0.74%
optimised_risk_parity	6.81%	11.24%	14.42%	-59.03%	\$698,771	\$2,318,481	\$5,315,155	0.40%
optimised_dividend_income	5.46%	11.06%	14.65%	-70.51%	\$480,064	\$2,206,944	\$5,648,305	1.32%
optimised_score_weighted	6.05%	11.00%	14.38%	-64.77%	\$564,972	\$2,173,462	\$5,263,575	0.82%
optimised_regime_aware	5.53%	10.84%	14.28%	-67.16%	\$489,389	\$2,084,931	\$5,124,773	0.94%
optimised_cvar_constrained	5.29%	10.13%	13.31%	-64.80%	\$457,268	\$1,723,804	\$3,992,373	0.82%
llm_benchmark	6.69%	9.69%	11.95%	-44.51%	\$675,422	\$1,529,558	\$2,796,904	0.30%
optimised_mean_variance	5.69%	9.60%	12.49%	-58.57%	\$511,192	\$1,494,769	\$3,218,606	0.56%
drl_baseline	5.46%	9.45%	12.36%	-59.98%	\$892,862,365	\$2,673,477,403	\$5,788,989,578	0.70%
final_portfolio	5.51%	9.43%	12.34%	-60.03%	\$904,708,919	\$2,654,953,471	\$5,761,497,823	0.74%
clean_sheet	5.33%	9.27%	12.26%	-58.97%	\$462,114	\$1,367,344	\$3,028,429	0.74%
drl_challenger_raw	5.37%	9.15%	11.94%	-58.09%	\$870,200,265	\$2,462,818,359	\$5,187,127,404	0.72%
trend_risk_controlled_indices	1.63%	4.58%	6.93%	-57.25%	\$160,960	\$375,251	\$721,469	2.08%

Monte Carlo uses correlated Student-t shocks, AR(1) persistence, and EWMA volatility to create fat-tailed, volatility-clustered paths. It is a modeled stress distribution, not a forecast promise. The lowest simulated 5th-percentile CAGR is trend_risk_controlled_indices at 1.63%.

Untouched Embargo Comparison



Portfolio	Sample	Months	cagr	Volatility	sharpe	Max drawdown
clean_sheet	development	318	8.72%	10.89%	0.65	-27.73%
clean_sheet	untouched_embargo	36	11.19%	8.90%	0.75	-4.47%
current_portfolio	development	318	13.94%	16.58%	0.76	-42.57%
current_portfolio	untouched_embargo	36	15.09%	17.65%	0.64	-13.50%
drl_baseline	development	318	8.56%	11.19%	0.62	-28.45%
drl_baseline	untouched_embargo	36	13.94%	10.41%	0.89	-6.75%
drl_challenger_raw	development	318	8.29%	10.66%	0.62	-27.87%
drl_challenger_raw	untouched_embargo	36	13.74%	9.83%	0.92	-6.36%
final_portfolio	development	318	8.56%	11.19%	0.62	-28.45%
final_portfolio	untouched_embargo	36	13.94%	10.41%	0.89	-6.75%
llm_benchmark	development	318	9.47%	9.07%	0.83	-20.91%
llm_benchmark	untouched_embargo	36	9.72%	9.20%	0.58	-8.37%
optimised_cvar_constrained	development	318	9.15%	12.28%	0.62	-32.27%
optimised_cvar_constrained	untouched_embargo	36	14.89%	11.08%	0.92	-7.55%
optimised_dividend_income	development	318	10.03%	14.36%	0.60	-41.29%
optimised_dividend_income	untouched_embargo	36	14.82%	11.90%	0.86	-8.73%
optimised_mean_variance	development	318	8.68%	11.07%	0.64	-29.60%
optimised_mean_variance	untouched_embargo	36	14.49%	10.50%	0.93	-7.24%
optimised_regime_aware	development	318	9.82%	13.24%	0.63	-36.59%
optimised_regime_aware	untouched_embargo	36	15.49%	11.38%	0.94	-7.47%
optimised_risk_parity	development	318	10.69%	11.58%	0.77	-30.56%
optimised_risk_parity	untouched_embargo	36	12.67%	11.30%	0.73	-8.71%

Portfolio	Sample	Months	cagr	Volatility	sharpe	Max drawdown
optimised_score_weighted	development	318	10.02%	12.84%	0.66	-35.52%
optimised_score_weighted	untouched_embargo	36	15.51%	11.42%	0.94	-7.29%
portfolio_aware_overlay	development	318	12.51%	14.60%	0.75	-41.21%
portfolio_aware_overlay	untouched_embargo	36	17.00%	12.95%	0.94	-11.35%
trend_risk_controlled_indices	development	318	3.87%	8.61%	0.26	-23.30%
trend_risk_controlled_indices	untouched_embargo	36	9.48%	8.63%	0.59	-8.26%

The embargo table compares the development sample with the final untouched 36 months. A sharp fall in Sharpe or return signals instability; agreement is encouraging but cannot cure retrospective selection bias.

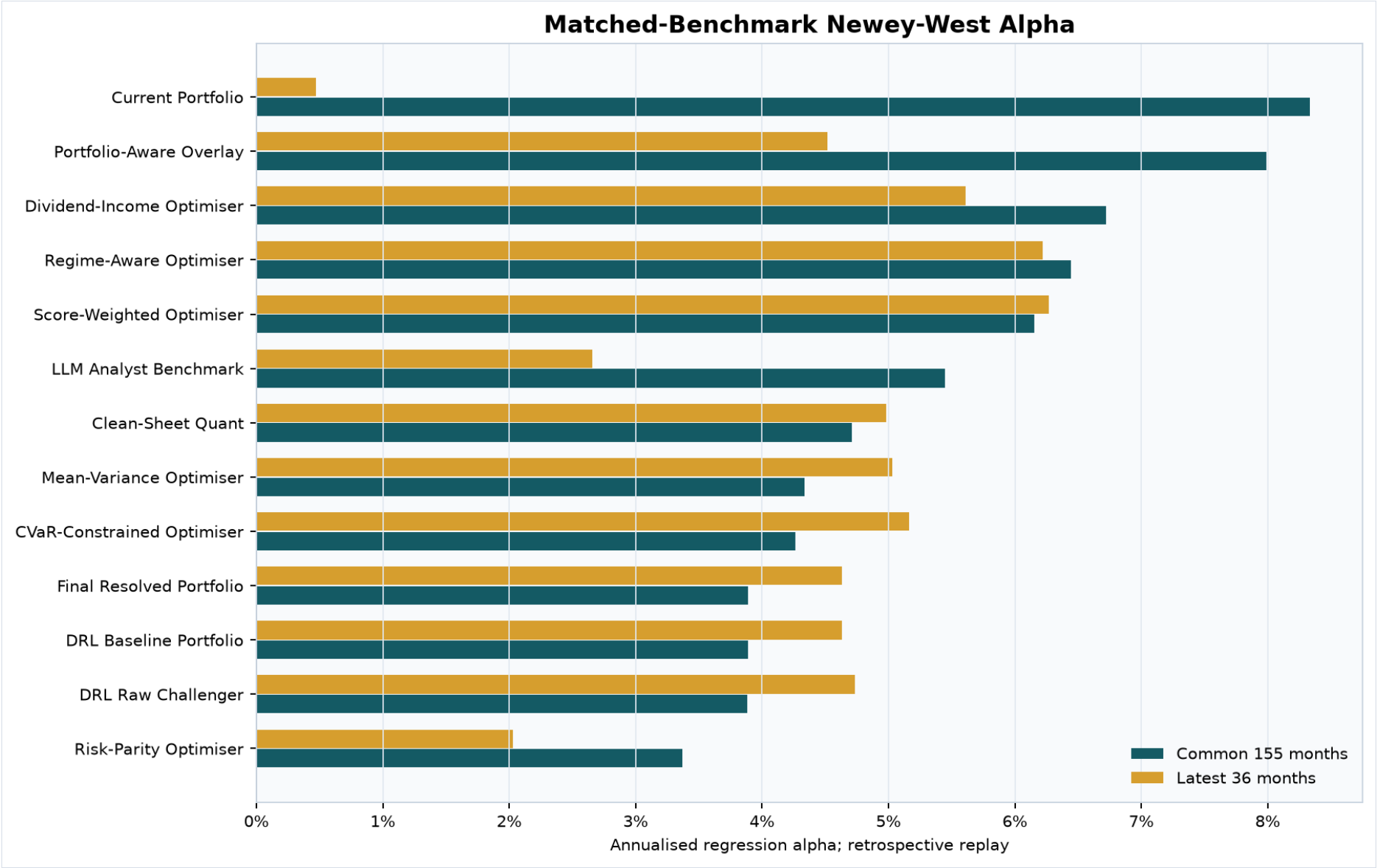
Statistical Significance and Multiple Testing

Portfolio	annualised_sharpe	PSR	MinTRL months	sidak_significant	DSR	cluster_adjusted_deflated_sharpe_ratio
llm_benchmark	0.81	100.00%	53.898	True	99.92%	99.99%
portfolio_aware_overlay	0.77	100.00%	62.056	True	99.80%	99.97%
optimised_risk_parity	0.77	100.00%	62.373	True	99.80%	99.97%
current_portfolio	0.75	99.99%	64.858	True	99.73%	99.96%
optimised_score_weighted	0.69	99.99%	73.165	True	99.43%	99.90%
clean_sheet	0.66	99.98%	75.886	True	99.26%	99.87%
optimised_regime_aware	0.66	99.98%	78.304	True	99.18%	99.85%
optimised_mean_variance	0.66	99.98%	79.627	True	99.16%	99.84%
drl_challenger_raw	0.65	99.97%	82.858	True	98.98%	99.80%
final_portfolio	0.65	99.97%	83.862	True	98.92%	99.79%

Portfolio	annualised_sharpe	PSR	MinTRL months	sidak_significant	DSR	cluster_adjusted_deflated_sharpe_ratio
drl_baseline	0.65	99.97%	83.862	True	98.92%	99.79%
optimised_cvar_constrained	0.65	99.96%	84.416	True	98.91%	99.78%
optimised_dividend_income	0.63	99.96%	85.356	True	98.75%	99.76%
trend_risk_controlled_indices	0.30	93.87%	401.646	False	67.58%	84.38%

PSR and MinTRL evaluate reliability, Sidak controls family-wise error across tested strategies, and DSR penalizes multiple trials and non-normal returns. 13 of 14 results clear Sidak; significance does not remove retrospective selection look-ahead.

Alpha and Strategy-Selection Overfitting



observations	Raw trials	Unique trials	Global max-t p-value	CSCV PBO	Median selected IS IR	Median selected OOS IR	Selection IR haircut	Selection risk	Deployable alpha
155	13	12	0.0004	19.91%	1.32	0.66	50.29%	LOW	NOT_ESTABLISHED

Portfolio	annualised_active_return	information_ratio	Max-t p-value	FWER significant	Selected in CSCV	Duplicate path
Portfolio-Aware Overlay	6.96%	1.04	0.0004	True	48.38%	
LLM Analyst Benchmark	5.11%	0.87	0.0054	True	32.36%	
Dividend-Income Optimiser	6.10%	0.84	0.0068	True	17.42%	
Current Portfolio	7.50%	0.83	0.0076	True	1.62%	
Regime-Aware Optimiser	5.42%	0.71	0.0262	True	0.00%	
Score-Weighted Optimiser	5.11%	0.70	0.0274	True	0.00%	
Clean-Sheet Quant	3.99%	0.55	0.1050	False	0.22%	
CVaR-Constrained Optimiser	3.00%	0.40	0.2791	False	0.00%	
Mean-Variance Optimiser	2.84%	0.38	0.3081	False	0.00%	
DRL Raw Challenger	2.38%	0.33	0.3919	False	0.00%	
Final Resolved Portfolio	2.28%	0.30	0.4509	False	0.00%	
DRL Baseline Portfolio	2.28%	0.30	0.4509	False	0.00%	final_portfolio
Risk-Parity Optimiser	1.65%	0.24	0.5755	False	0.00%	

Portfolio	Sample	observations	annualised_alpha	beta	NW t-stat	NW p-value	Evidence status
Current Portfolio	common_investable_window	155	8.33%	0.84	3.397	0.0007	RETROSPECTIVE_ONLY
Portfolio-Aware Overlay	common_investable_window	155	7.99%	0.80	5.005	0.0000	RETROSPECTIVE_ONLY
Dividend-Income Optimiser	common_investable_window	155	6.72%	0.86	3.678	0.0002	RETROSPECTIVE_ONLY

Portfolio	Sample	observations	annualised_alpha	beta	NW t-stat	NW p-value	Evidence status
Regime-Aware Optimiser	common_investable_window	155	6.44%	0.82	3.701	0.0002	RETROSPECTIVE_ONLY
Score-Weighted Optimiser	common_investable_window	155	6.16%	0.79	3.969	0.0001	RETROSPECTIVE_ONLY
LLM Analyst Benchmark	common_investable_window	155	5.45%	0.87	3.326	0.0009	RETROSPECTIVE_ONLY
Clean-Sheet Quant	common_investable_window	155	4.71%	0.79	2.591	0.0096	RETROSPECTIVE_ONLY
Mean-Variance Optimiser	common_investable_window	155	4.34%	0.72	2.803	0.0051	RETROSPECTIVE_ONLY
CVaR-Constrained Optimiser	common_investable_window	155	4.26%	0.77	2.836	0.0046	RETROSPECTIVE_ONLY
Final Resolved Portfolio	common_investable_window	155	3.89%	0.71	2.651	0.0080	RETROSPECTIVE_ONLY
DRL Baseline Portfolio	common_investable_window	155	3.89%	0.71	2.651	0.0080	RETROSPECTIVE_ONLY
DRL Raw Challenger	common_investable_window	155	3.88%	0.71	2.757	0.0058	RETROSPECTIVE_ONLY
Risk-Parity Optimiser	common_investable_window	155	3.37%	0.75	2.289	0.0221	RETROSPECTIVE_ONLY
Score-Weighted Optimiser	last_36_months	36	6.27%	0.57	2.005	0.0449	RETROSPECTIVE_ONLY
Regime-Aware Optimiser	last_36_months	36	6.22%	0.55	1.879	0.0602	RETROSPECTIVE_ONLY
Dividend-Income Optimiser	last_36_months	36	5.61%	0.62	1.625	0.1043	RETROSPECTIVE_ONLY
CVaR-Constrained Optimiser	last_36_months	36	5.17%	0.60	1.550	0.1211	RETROSPECTIVE_ONLY
Mean-Variance Optimiser	last_36_months	36	5.03%	0.57	1.577	0.1149	RETROSPECTIVE_ONLY
Clean-Sheet Quant	last_36_months	36	4.98%	0.37	1.322	0.1861	RETROSPECTIVE_ONLY
DRL Raw Challenger	last_36_months	36	4.73%	0.54	1.544	0.1226	RETROSPECTIVE_ONLY
Final Resolved Portfolio	last_36_months	36	4.63%	0.55	1.454	0.1460	RETROSPECTIVE_ONLY
DRL Baseline Portfolio	last_36_months	36	4.63%	0.55	1.454	0.1460	RETROSPECTIVE_ONLY
Portfolio-Aware Overlay	last_36_months	36	4.51%	0.85	1.098	0.2721	RETROSPECTIVE_ONLY
LLM Analyst Benchmark	last_36_months	36	2.66%	0.60	0.618	0.5367	RETROSPECTIVE_ONLY
Risk-Parity Optimiser	last_36_months	36	2.03%	0.66	0.645	0.5190	RETROSPECTIVE_ONLY

Portfolio	Sample	observations	annualised_alpha	beta	NW t-stat	NW p-value	Evidence status
Current Portfolio	last_36_months	36	0.47%	1.06	0.077	0.9386	RETROSPECTIVE_ONLY

Newey-West regressions test alpha against each portfolio-specific regional index blend while allowing serial correlation. The block-bootstrap max-t test controls selection across the whole strategy family, and CSCV repeatedly selects in one half of the history and ranks that winner in the other half. 6 unique results clear the max-t family-wise test. CSCV estimates PBO at 19.91%, while the median selected information ratio falls from 1.32 in-sample to 0.66 out-of-sample. The largest measured common-window alpha is 8.33% for Current Portfolio. Every long portfolio path is still a retrospective holdings replay, so these tests measure path robustness, not deployable stock-selection alpha.

Data Quality and Causality

Execution and Liquidity

Portfolio	Total modeled cost	Annual turnover	Max ADV participation	Constrained trade events	Max unfilled target
current_portfolio	\$188,405,153	17.52%	5.00%	1645	27.56%
portfolio_aware_overlay	\$167,507,440	23.94%	5.00%	1714	25.32%
final_portfolio	\$83,637,346	21.63%	5.00%	2332	37.65%
drl_baseline	\$83,637,346	21.63%	5.00%	2332	37.65%
drl_challenger_raw	\$78,761,389	21.62%	5.00%	3218	33.86%
optimised_risk_parity	\$61,615	27.98%	5.00%	1	0.81%
optimised_dividend_income	\$52,785	30.88%	5.00%	10	0.53%
optimised_score_weighted	\$52,306	28.98%	5.00%	10	0.47%
optimised_regime_aware	\$49,074	29.43%	5.00%	10	0.46%
llm_benchmark	\$45,544	24.15%	5.00%	61	4.42%
optimised_cvar_constrained	\$45,242	29.32%	5.00%	2	0.91%

Portfolio	Total modeled cost	Annual turnover	Max ADV participation	Constrained trade events	Max unfilled target
optimised_mean_variance	\$40,686	27.13%	5.00%	5	4.42%
clean_sheet	\$39,945	25.90%	5.00%	11	4.04%
trend_risk_controlled_indices	\$28,265	126.60%		0	0.00%

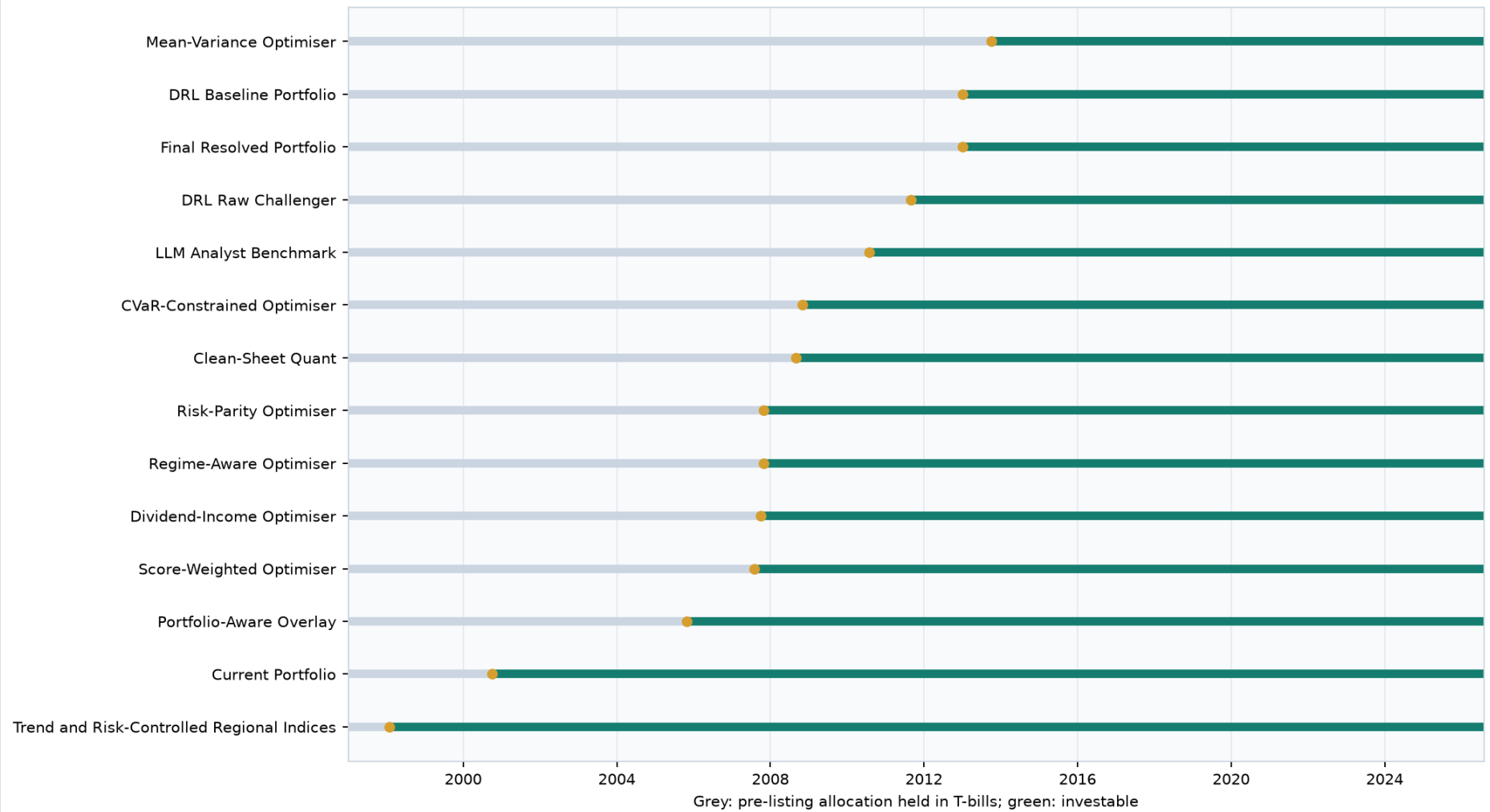
Execution rows show dollar costs, turnover, peak ADV participation, constrained trades, and unfilled weight. Large current-derived AUM can leave low-liquidity targets partly in cash, making these results capacity-aware.

Adjusted-Price Repairs

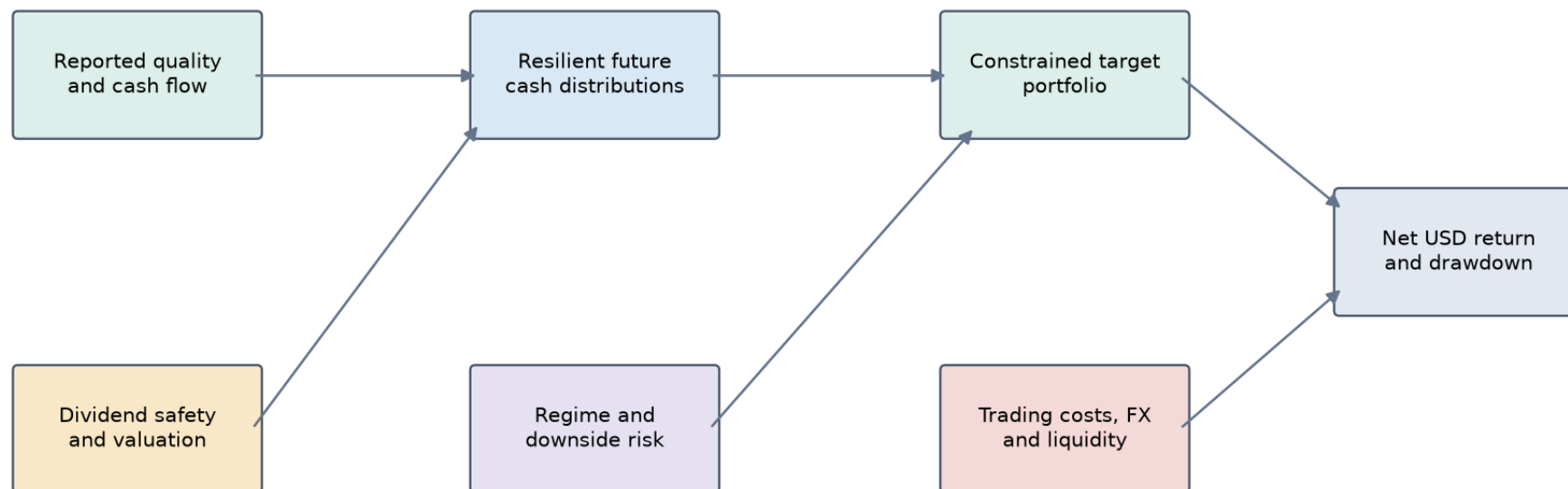
symbol	Event date	Auditable repair	Raw return	Post-repair return	Scale factor
600018.SS	2006-10-26 00:00:00	persistent_level_shift_historical_rescale	-76.85%	0.00%	0.232
GEBN.SW	1999-06-22 00:00:00	persistent_level_shift_historical_rescale	-58.96%	0.00%	0.410
SHEL.L	1997-07-01 00:00:00	persistent_level_shift_historical_rescale	-71.74%	0.00%	0.283
TTE.PA	2005-12-26 00:00:00	isolated_spike_log_interpolation	300.00%	-0.16%	0.250
ULVR.L	1997-08-26 00:00:00	persistent_level_shift_historical_rescale	322.53%	0.00%	4.225
ULVR.L	1997-10-13 00:00:00	persistent_level_shift_historical_rescale	83.68%	0.00%	1.837

The adjusted-price process made 6 explicit repairs. Each row records the raw move, rule, repaired return, and scale factor so provider anomalies remain auditable.

When at Least 80% of Each Intended Asset Allocation Was Tradable



Causal Theory Tested by the Model and Degraded by Execution Frictions



Bias and Limitation Register

category	severity	applies_to	status	treatment
selection look-ahead	CRITICAL	all current portfolio-output replays	ACKNOWLEDGED	Labelled retrospective holdings replay; excluded from claims of 1997 model selection skill.
survivorship bias	CRITICAL	current selected securities	ACKNOWLEDGED	Pre-listing capital stays in T-bills; current holdings are never presented as a historical universe.
	HIGH	full stock-selection strategy	SEPARATED	

category	severity	applies_to	status	treatment
point-in-time model evidence				The 25-month reconstructed decision history is reported separately and never spliced into the long replay.
corporate actions	MEDIUM	holding histories	MODELED	Yahoo adjusted closes include splits and distributions where provider history supports them.
currency conversion	MEDIUM	non-USD holdings and indices	MODELED	Historical FRED FX converts local adjusted closes to USD; ECU bridges pre-euro observations.
benchmark dividend consistency	MEDIUM	regional price indices	WARNING	DAX is a performance index; other regional index series may omit dividends. SPY is included as a total-return proxy.
transaction costs and liquidity	MEDIUM	portfolio outputs and index challenger	MODELED	Commission, spread, slippage, square-root impact, missing-ADV penalty, and 5% ADV caps are reported.
annual bank AUM charge	MEDIUM	portfolio outputs and index challenger	MODELED	A 25 bp charge is deducted once each December from then-current AUM; external benchmarks remain uncharged references.
conditional regime analysis	MEDIUM	rate, market, and recession tables	DESCRIPTIVE	Rate and market inputs are lagged one month; NBER recession labels are retrospective and are not presented as tradable signals.
macro-event windows	MEDIUM	event comparisons and shaded plots	DESCRIPTIVE	Configured market-response windows overlap monthly returns and do not assert legal conflict dates or isolate causal effects.
short-sale constraints	LOW	all tested portfolios	NOT_APPLICABLE	Every tested allocation is long-only; no borrow availability or fee assumption is required.
multiple testing	HIGH	portfolio comparison	CONTROLLED	Sidak, Deflated Sharpe, block-bootstrap max-t, duplicate-trial removal, and CSCV PBO are reported.
deployable alpha	CRITICAL	strategy-selection claims	NOT_ESTABLISHED	Retrospective alpha is diagnostic only; approval requires native point-in-time outperformance with sufficient history.
non-stationarity	HIGH	all historical and simulated evidence	RESIDUAL_RISK	Embargo, subperiod plots, block resampling, fat tails, and EWMA volatility expose but cannot remove structural breaks.

The limitation register distinguishes modeled, controlled, separated, and unresolved risks. The critical issue is that current holdings survived long enough to be selected today, so the 1997 replay cannot prove selection skill.

Point-in-Time Model Evidence

The repository's reconstructed model walk-forward is not spliced into the 1997 holdings replay. It remains a separate, shorter evidence set because it uses dated model decisions and realised outcomes.

strategy	observations	annualised_return	annualised_volatility	sharpe	maximum_drawdown	annualised_cost_drag	evidence_mode
wolf_cvar	25	19.39%	10.51%	1.85	-5.51%	1.97%	reconstructed_pit_proxy
equal_weight_eligible	25	24.99%	11.29%	2.21	-5.17%	0.64%	reconstructed_pit_proxy
cap_weight_eligible	25	17.62%	8.88%	1.98	-3.99%	0.57%	reconstructed_pit_proxy

Point-in-Time Alpha Tests

strategy	benchmark	observations	annualised_active_return	Regression alpha	NW t-stat	NW p-value	Extra cost drag	Alpha verdict	Deployable alpha
wolf_cvar	cap_weight_eligible	25	1.66%	3.37%	0.559	0.5763	1.40%	INSUFFICIENT_HISTORY	NOT_ESTABLISHED
wolf_cvar	equal_weight_eligible	25	-4.75%	4.82%	0.770	0.4414	1.34%	INSUFFICIENT_HISTORY	NOT_ESTABLISHED

The point-in-time table contains 25 dated decision months and remains separate from the long holdings replay. It is the more relevant evidence for decisions made with contemporaneous data. Against equal-weight eligible stocks, annualised active return is -4.75% and the alpha verdict is INSUFFICIENT_HISTORY. Against cap-weight eligible stocks, annualised active return is 1.66% and the verdict is INSUFFICIENT_HISTORY. Native live evidence and at least 60 months are required before alpha can be considered deployable.

Research evidence only. The paper used as the methodological specification is *The Three Types of Backtests* (Joubert, Sestovic, Barziy, Distaso, and Lopez de Prado, 29 July 2024). Raw Yahoo Finance and FRED histories remain outside Git; compact outputs, manifests, and checksums are included.